

Zeus: Towards Tuning-Free Foundation Model for Time Series Analysis

Yisong Fu^{1,2} Zezhi Shao¹ Chengqing Yu¹ Yujie Li^{1,2} Yongjun Xu^{1,2,3} Xueqi Cheng^{1,2} Fei Wang^{1,2}

Abstract

We present ZEUS, a unified tuning-free Time Series Foundation Model (TSFM) that delivers superior performance across diverse analysis tasks without any task-specific fine-tuning. Unlike prior studies that primarily focus on zero-shot forecasting but require task-specific tuning for other tasks, ZEUS bridges this gap by addressing two fundamental challenges in multi-task generalization. First, to reconcile point-level granularity with long-sequence scalability, ZEUS incorporates a multi-scale Transformer featuring point-wise tokenization and a U-shaped hierarchy, effectively balancing fine-grained fidelity with computational efficiency. Second, to accommodate varying inductive biases across different tasks, ZEUS introduces Multi-Objective Temporal Masking (MOTM), a unified strategy that supports heterogeneous tasks (*e.g.*, extrapolation, interpolation, and global abstraction) within a single framework. Extensive experiments across five representative tasks demonstrate that ZEUS consistently achieves competitive results in tuning-free settings, underscoring its potential as a general-purpose TSFM. The code is available at <https://github.com/GestaltCogTeam/Zeus>.

1. Introduction

Time series analysis is a pivotal field with broad-ranging applications, from weather forecasting (Fu et al., 2025) and physiological anomaly detection (Šabić et al., 2021), to data imputation (Luo et al., 2018) and human activity recognition (Vrigkas et al., 2015). Its profound practical utility continues to attract significant research and industrial interest.

¹State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences ²University of Chinese Academy of Sciences ³Xiamen Institute of Data Intelligence. Yisong Fu <fuyisong24s@ict.ac.cn>. Correspondence to: Fei Wang <wangfei@ict.ac.cn>.

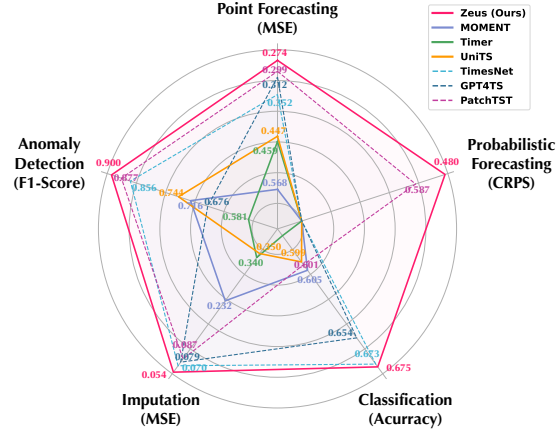


Figure 1. Overall performance comparison of ZEUS under the tuning-free setting. ZEUS surpasses full-shot task-specific models (*dashed lines*) and significantly outperforms other TSFMs in tuning-free setting (*solid lines*).

Inspired by the success of foundation models in language (OpenAI, 2023), images (Ramesh et al., 2021), and videos (Liu et al., 2024c), researchers have been striving to develop general-purpose time series foundation models (TSFMs). Trailblazing efforts like MOMENT (Goswami et al., 2024), Timer (Liu et al., 2024b), and UniTS (Gao et al., 2024) have shown the potential of unified architectures pretrained on large-scale datasets to address multiple downstream tasks.

Despite these advances, such models have primarily demonstrated zero-shot capability only in forecasting tasks, while adapting to other downstream tasks typically requires additional training (Liu et al., 2024b; Gao et al., 2024). This limitation contrasts with the general expectations of foundation models and has consequently led most recent efforts to focus predominantly on forecasting (Shi et al., 2025; Cohen et al., 2025; Auer et al., 2025). This limitation can be traced to two fundamental dilemmas in existing TSFMs.

❶ Architectural dilemma between granularity and scalability. Most existing TSFMs adopt patch-wise tokenization (Nie et al., 2023) to improve semantic density and computational efficiency. However, this design inevitably sacrifices point-level temporal details, hindering its effectiveness on reconstruction-based tasks like imputation and anomaly detection. For example, MOMENT, pretrained with patch-level reconstruction, exhibits a significant performance drop

when shifting from patch-missing to point-missing imputation (evidenced in Table 6). In contrast, point-wise tokenization (Ansari et al., 2024; Shi et al., 2025) preserves fine-grained structure but suffers from low information density and prohibitive computational overhead on long sequences.

② Training dilemma for divergent inductive biases. Although recent works attempt a unified modeling paradigm, they often neglect the fundamentally different inductive biases required by each objective: forecasting relies on extrapolation, imputation and anomaly detection necessitate interpolation, and classification requires global abstraction. This heterogeneity indicates that a single, monolithic training objective, such as BERT-style masked reconstruction (Goswami et al., 2024; Zhang et al., 2025) or GPT-style autoregressive generation (Liu et al., 2024b), is unlikely to simultaneously endow all necessary capabilities.

Collectively, these dilemmas have hindered prior TSFMs from fully generalizing across diverse tasks, often necessitating task-specific fine-tuning to achieve strong performance. To address these challenges, we introduce ZEUS, a unified TSFM that bridges point-level fidelity and long-sequence scalability while supporting diverse task-specific inductive biases within a single pretraining framework. ZEUS is designed to operate in a fully tuning-free¹ manner, enabling out-of-box deployment across diverse downstream tasks.

To overcome the tension between granularity and scalability, ZEUS adopts a U-shaped multi-scale hierarchy with point-wise tokenization. It follows a *fine-to-coarse-to-fine* information flow, progressively aggregating fine-grained information into coarse-grained latent representations and then symmetrically refining them. Lightweight Transformer blocks are used at high resolutions to preserve local details, while deeper and wider blocks operate on compressed representations to efficiently capture global dependencies.

Complementing this architecture, we address the challenge of divergent inductive biases with MOTM, a Multi-Objective Temporal Masking strategy that jointly trains ZEUS for extrapolation, interpolation, and global abstraction. By exposing the model to predictive, point-wise, and block-wise corruption patterns during pretraining, MOTM enables ZEUS to learn a unified yet versatile representation space that supports out-of-the-box performance across diverse downstream tasks.

Experimentally, ZEUS achieves the consistent state-of-the-art performance in a tuning-free manner across widely recognized benchmarks for five downstream tasks (Figure 1), including long time series benchmark (Wu et al., 2023) for point forecasting and imputation, GIFT-Eval (Aksu et al., 2024) for probabilistic forecasting, UCR anomaly archive

(Wu & Keogh, 2021) for anomaly detection, and UEA archive for classification. These results demonstrate the strong generalizability of ZEUS across diverse tasks. In summary, our contribution is three-fold:

- We present ZEUS, a unified multi-scale Transformer that preserves point-level fidelity while efficiently modeling high-level semantics over long sequences.
- We introduce MOTM, a multi-objective temporal masking strategy that aligns diverse task-specific inductive biases within a single pretraining framework.
- To the best of our knowledge, ZEUS is the first TSFM to achieve competitive performance across five downstream tasks without task-specific adaptation, as validated on well-established benchmarks.

2. Related Work

2.1. Task-Specific Time Series Model

Time series analysis plays a fundamental role in a wide range of applications, including forecasting, imputation, anomaly detection, and classification. Early statistical and traditional machine learning methods typically rely on case-by-case fitting, facing limitations when applied to large-scale data. With the advancement of deep learning, task-specific neural models have become the dominant paradigm (Xu et al., 2021; Huang et al., 2025). CNN-based models, such as TS2Vec (Yue et al., 2022), TimesNet (Wu et al., 2023) and ModernTCN (donghao & xue, 2024), have shown strong potential to support multiple tasks within a unified framework. Transformer variants (Wu et al., 2021; Eldele et al., 2024; Nie et al., 2023; Liu et al., 2024a) further enhance the ability to capture long-range dependencies and complex temporal patterns. Inspired by advances in LLMs, recent work has explored transferring pretrained language models to time series tasks (Gruver et al., 2023; Zhou et al., 2023; Jin et al., 2024). Typically, GPT4TS (Zhou et al., 2023) explores the use of pretrained GPT-2 as a frozen backbone for time series analysis, adapting it to downstream tasks through lightweight output projections. However, these methods still rely on task-specific training for each downstream application.

2.2. Time Series Foundation Models

Building upon task-specific precursors, recent research has increasingly explored pretraining foundation models on large-scale time series corpora to enhance cross-task generalization, with Transformer (Vaswani et al., 2017) becoming the predominant architecture. Representative approaches such as MOMENT (Goswami et al., 2024) and TimesBERT (Zhang et al., 2025) adopt BERT-style masked reconstruction objectives to learn contextualized representations by

¹In this paper, the term *tuning-free* refers to performing inference without tuning or retraining any model parameters.

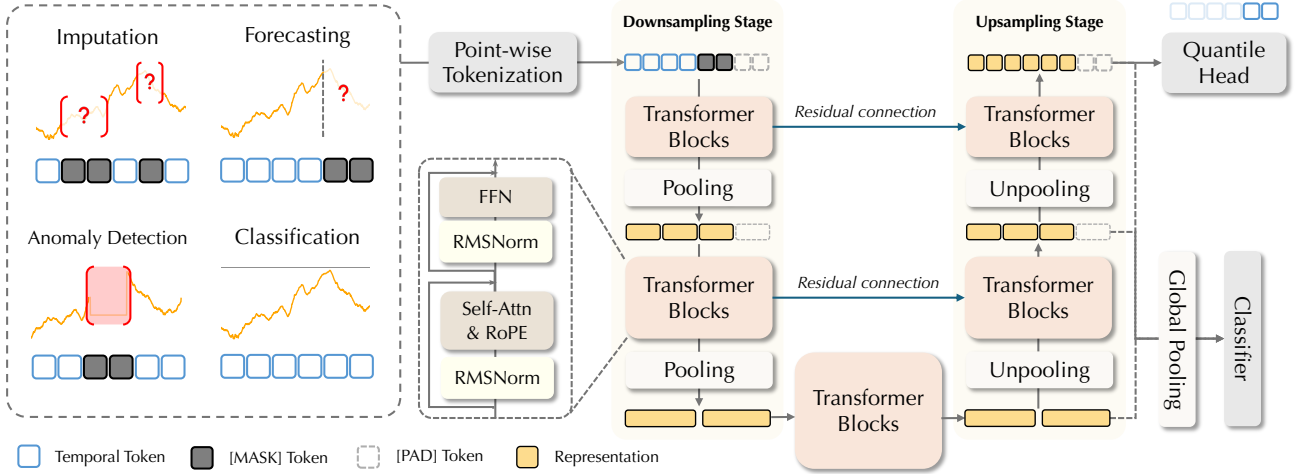


Figure 2. Overall architecture of ZEUS. Inputs from different downstream tasks are first unified into a common format and converted into point-wise tokens via tokenization. The resulting sequence is then processed by a U-shaped multi-scale Transformer. Quantile head is used to produce probabilistic outputs, while for classification tasks, global pooling is applied to obtain sequence-level representations.

recovering masked patches. UniTS (Gao et al., 2024) extends this paradigm by incorporating task prompts, facilitating flexible task adaptation. In contrast, Timer (Liu et al., 2024b) adopts a GPT-style autoregressive formulation to cast diverse time series tasks into a unified generative framework. Despite their progress, existing TSFMs remain constrained by architectural and training-design limitations, which hinder fine-grained modeling and necessitate task-specific adaptation in practice.

3. Zeus

3.1. Overall Framework

ZEUS is a multi-scale encoder-only Transformer architecture for general-purpose time series analysis. As illustrated in Figure 2, it adopts point-wise tokenization to preserve fine-grained temporal resolution, and processes the resulting tokens through a symmetric downsampling–upsampling hierarchy of Transformer encoders. Representations are progressively aggregated from fine to coarse scales and then refined back to fine resolution. This design resolves the dilemma between granularity and scalability: it preserves point-level details required by reconstruction-oriented tasks while promoting semantically compact and computationally efficient representations for large-scale pretraining.

3.2. Tokenization

ZEUS adopts a point-wise tokenization scheme, where each time step is treated as an individual token. We conduct pretraining in a univariate paradigm and employ the channel-independent strategy to handle multivariate time series (Nie et al., 2023). Given a time series $\mathbf{x} = \{x_1, x_2, \dots, x_T\}$, we first apply instance normalization (Kim et al., 2021) to

remove scale variations. Then each time step is mapped to a hidden representation via a gated embedding layer:

$$\mathbf{h}_t^{(0)} = \mathbf{W}_r x_t + \mathbf{W}_d (\sigma(\mathbf{W}_g x_t) \odot \mathbf{W}_u x_t), \quad (1)$$

where $\sigma(\cdot)$ denotes a nonlinear activation. This design increases the expressive capacity of token embeddings.

We further introduce two learnable tokens: a [MASK] token for masked reconstruction and a [PAD] token for variable-length sequences and multi-scale alignment, enabling unified handling of masking, missing values, and padding. As shown in Figure 2, by replacing temporal tokens with the [MASK] token, diverse downstream tasks can be uniformly formulated under a masked modeling paradigm, allowing ZEUS to support multiple tasks within a unified framework. In forecasting tasks, several [MASK] tokens are appended after the historical window; in imputation tasks, missing values are replaced with [MASK]; in anomaly detection tasks, the target segment can be replaced with [MASK], and reconstruction or prediction errors are used as anomaly scores.

3.3. Multi-Scale Architecture

While point-wise tokenization preserves temporal fidelity, it introduces substantial computational challenges for long sequences. ZEUS resolves this tension through a U-shaped hierarchy that progressively adjusts temporal resolution across symmetric scales $\{s_1, s_2, \dots, s_K\}$, where $s_i = s_{K-i+1}$ defines the number of time steps aggregated into a token.

Downsampling and Upsampling Stages This U-shaped architecture consists of a downsampling stage to compress fine-grained information into high-level semantics via pooling operation, and a symmetric upsampling stage, which

employs unpooling to progressively restore local details.

In the downsampling stage, given the hidden states $\mathbf{h}^{(i)} \in \mathbb{R}^{T_i \times d_i}$ at scale $i < \frac{K-1}{2}$, a pooling layer aggregates $r = s_{i+1}/s_i$ adjacent tokens into a single representation. The pooled representations are then processed by a Transformer encoder to model higher-level temporal dependencies.

$$\mathbf{p}^{(i)} = \text{Reshape}(\mathbf{h}^{(i)}, \mathbb{R}^{\frac{L_i}{r} \times (rd_i)}) \mathbf{W}_p, \quad (2)$$

$$\mathbf{h}^{(i+1)} = \text{TrfmEncoder}(\mathbf{p}^{(i)}), \quad (3)$$

where $\mathbf{W}_p \in \mathbb{R}^{(rd_i) \times d_{i+1}}$ is a learnable linear projection.

Conversely, in the upsampling stages ($i \geq \frac{K-1}{2}$), an unpooling layer expands the sequence and integrates high-resolution features from the corresponding scale via residual skip connections:

$$\mathbf{P}^{(i)} = \text{Reshape}(\mathbf{h}^{(i)} \mathbf{W}_u, \mathbb{R}^{(rL_i) \times d_{i+1}}) + \mathbf{h}^{(K-i+1)}. \quad (4)$$

The resulting representations are then fed into a Transformer encoder to refine fine-grained temporal details. In this architecture, lightweight Transformer blocks operate at fine scales to capture local patterns, while deeper and wider blocks process coarser representations to model long-range dependencies. This multiscale design enables ZEUS to efficiently balance temporal fidelity and scalability.

Transformer Block Each Transformer block employs multi-head self-attention with rotary positional embeddings (Su et al., 2024), together with gated feed-forward networks (Shazeer, 2020). To enhance training stability, we adopt RMSNorm (Zhang & Sennrich, 2019) and a pre-LN scheme (Xiong et al., 2020). All attention modules are implemented with FlashAttention v2 (Dao, 2024), enabling memory-efficient and scalable training on long sequences.

Quantile Head ZEUS employs a quantile head that provides $|Q|$ quantile values for each time step, enabling probabilistic reconstruction instead of point estimation. Formally, the head implements a mapping $\mathbb{R}^{d_K} \rightarrow \mathbb{R}^{|Q|}$. In our implementation, ZEUS predicts a set of nine quantile levels $Q = \{0.1, 0.2, \dots, 0.9\}$. For classification task, we directly use the globally pooled representations.

3.4. Multi-Objective Temporal Masking

The key challenge for TSFMs to generalize across tasks lies in the distinct inductive biases required by different objectives. Forecasting requires extrapolation, whereas imputation relies on interpolation. Anomaly detection and classification introduce more complex demands: point anomalies require sensitivity to local variations, while contextual

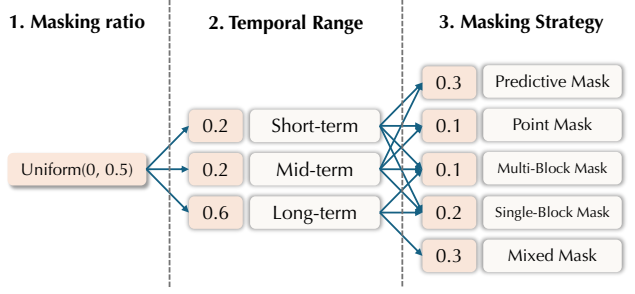


Figure 3. The MOTM pipeline. MOTM hierarchically determines the masking ratio, scales the temporal scope, and applies diverse masking strategies to jointly optimize for extrapolation, interpolation, and local-global feature extraction.

anomalies necessitate modeling global consistency (Liu et al., 2025a). Moreover, classification calls for both global abstraction and the identification of local shapelets (Le et al., 2022). However, current TSFMs typically employ a simple BERT-style (Goswami et al., 2024; Zhang et al., 2025) or GPT-style objective (Liu et al., 2024b), which fails to simultaneously capture these heterogeneous inductive biases.

To bridge this gap, we introduce multi-objective temporal masking (MOTM), a unified masking-based training strategy that equips ZEUS with heterogeneous inductive biases. Figure 3 illustrates the pipeline of MOTM. For each instance, the overall corruption ratio is sampled from $\mathcal{U}(0, 0.5)$ to enhance robustness to varying levels of missing. We then select a temporal scope that jointly cover short-, mid-, and long-term dependencies. Finally, a masking scheme is sampled from a diverse set, including predictive, point, multi-block, and single-block masking, as well as their combinations. Below, we detail each masking strategy and discuss its role in shaping task-relevant inductive biases.

Predictive Mask To bolster extrapolative capacity essential for forecasting, we adopt a predictive masking scheme that masks a suffix of the sequence. Specifically, given a probability p , we mask the last $L_p = \lfloor Tp \rfloor$ time steps, requiring the model to learn temporal causality and capture long-range dependencies beyond the observed horizon.

Point Mask This widely-used strategy targets point-wise interpolation by randomly masking individual time steps. This fine-grained corruption forces the model to leverage local continuity and short-range correlations, while serving as a regularizer against overfitting to specific patterns.

Multi-Block Mask To simulate the contiguous missingness prevalent in real-world deployments, we introduce a multi-block masking strategy. Inspired by span-based corruption used in language modeling (Joshi et al., 2020; Lewis et al., 2020), this method shifts the objective from point-

Table 1. Zero-shot point forecasting results, averaged over four prediction lengths {96, 192, 336, 720}. Datasets used during pretraining are excluded from evaluation for the corresponding models and are denoted by a dash (—). Best and second-best results are shown in **bold** and underlined, respectively. Full results are provided in Table 7.

Models	Time Series Foundation Models (Zero-shot)								Pretrained Forecasting Models (Zero-shot)								Task-Specific Models (Supervised)									
	ZEUS (Ours)		MOMENT (2024)		Timer (2024b)		UniTS (2024)		Kairos _l (2025)		Toto _b (2025)		Sundial _l (2025b)		Time-MoE _l (2025)		ChronosBolt _b (2024)		ModernTCN (2024)		GPT4TS (2023)		TimesNet (2023)		PatchTST (2023)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.377	0.399	0.715	0.580	0.499	0.463	0.496	0.478	0.427	<u>0.410</u>	0.435	0.413	0.395	0.420	<u>0.394</u>	0.420	0.479	0.429	0.419	0.432	0.438	0.437	0.485	0.469	0.427	0.437
ETTh2	0.320	<u>0.364</u>	0.394	0.428	0.413	0.419	0.427	0.431	0.350	<u>0.374</u>	0.340	0.363	<u>0.334</u>	0.387	0.405	0.415	0.341	0.364	0.346	0.392	0.405	0.433	0.422	0.425	0.361	0.402
ETTm1	0.322	0.359	0.714	0.554	0.837	0.593	0.690	0.538	0.348	<u>0.365</u>	0.378	0.396	<u>0.331</u>	0.369	0.376	0.405	0.395	0.368	0.346	0.376	0.357	0.383	0.458	0.428	0.346	0.376
ETTm2	0.249	<u>0.305</u>	0.359	0.388	0.373	0.388	0.328	0.362	<u>0.252</u>	0.303	0.267	0.303	0.254	0.315	0.258	0.315	0.278	0.307	0.265	0.322	0.275	0.331	0.286	0.328	0.256	0.312
ECL	0.157	0.243	0.900	0.762	0.304	0.362	0.449	0.490	-	-	<u>0.161</u>	0.243	0.166	0.262	-	-	-	-	0.163	0.259	0.168	0.263	0.198	0.301	0.167	0.262
Weather	0.217	<u>0.247</u>	0.326	0.353	0.326	0.342	0.291	0.306	0.231	0.253	<u>0.224</u>	0.245	0.238	0.275	0.256	0.288	0.237	0.254	0.232	0.270	0.230	0.263	0.261	0.286	0.236	0.275
# Wins	19	14	0	0	0	0	0	0	<u>3</u>	1	0	<u>6</u>	0	0	0	0	1	5	1	0	0	0	0	0	0	0

wise reconstruction to structured recovery. However, instead of adopting the Poisson distribution commonly used in language modeling, we employ a uniform distribution, which poses a more challenging setting for time series. Specifically, given a masking budget $L_p = \lfloor Tp \rfloor$, we sample block lengths $\{\ell_k\}$ from $\ell_k \sim \mathcal{U}(1, 24)$, until $\sum_k \ell_k \approx L_p$, and randomly distribute the blocks along the sequence. This strategy compels the model to perform interpolation under structured missingness rather than simple local smoothing.

Single-Block Mask Complementary to multi-block masking, we introduce a single-block masking strategy that removes one long contiguous segment from an arbitrary position in the sequence. This strategy encourages the model to maintain global consistency, which is crucial for classification and contextual anomaly detection.

Mixed Mask We further employ a mixed masking scheme that combines multiple masking strategies to increase the difficulty of the pretraining objective. In practice, simpler masks (multi-block and point mask) are paired with harder ones (predictive and single-block mask), ensuring that mixed masking remains challenging yet learnable.

Training Objective Under all masking strategies, ZEUS is trained in a masked reconstruction manner with the quantile loss. Given a binary mask $\mathcal{M} \in \{0, 1\}^T$, the loss is computed only on masked positions:

$$\mathcal{L} = \frac{1}{|\mathcal{Q}||\mathcal{M}|} \sum_{t: \mathcal{M}_t=1} \sum_{q \in \mathcal{Q}} \begin{cases} q(y_t - \hat{y}_t^q), & \hat{y}_t^q \leq y_t, \\ (1-q)(\hat{y}_t^q - y_t), & \hat{y}_t^q > y_t. \end{cases} \quad (5)$$

Here, $\hat{y}_t^q$ denotes the prediction at quantile level q .

3.5. Pretraining Data

We train ZEUS on a large-scale corpus of real and synthetic time series, comprising about 300B observations collected from diverse domains and frequency. The real-world data are mainly sourced from the Chronos datasets (Ansari et al., 2024) and the GiftEvalPretrain datasets (Aksu et al., 2024). To enhance pattern diversity beyond real-world records, we additionally construct Aegis-Syn, a synthetic dataset that extends KernelSynth (Ansari et al., 2024) with richer temporal structures, particularly non-smooth and discontinuous patterns that are underrepresented by Gaussian process-based generators. To mitigate pattern imbalance in the pretraining corpus, we adopt a balanced sampling strategy (Shao et al., 2025a). All evaluation datasets are excluded from pretraining to prevent data leakage. Detailed descriptions of the pretraining data and Aegis-Syn are provided in Appendix E.

4. Experiments

Based on the above approach, we implemented and trained ZEUS with five scales and 12 Transformer layers, comprising approximately 100M parameters in total. Details are provided in Appendix A. In this section, we conduct a comprehensive evaluation on well-established benchmarks to assess ZEUS’s performance across five downstream tasks: point forecasting (§4.1), probabilistic forecasting (§4.2), imputation (§4.3), anomaly detection (§4.4), and classification (§4.5). See Appendix D for detailed experimental settings and baseline descriptions.

4.1. Point Forecasting

Setups We adopt six widely used long-term forecasting benchmarks (Wu et al., 2023; Shao et al., 2024) to evaluate ZEUS’s zero-shot performance in point forecast-

Table 2. Performance evaluation on the GIFT-Eval benchmark. Baseline results are officially reported by GIFT-Eval (Aksu et al., 2024).

Type	Pretrained Forecasting Models (Zero-Shot)										Supervised		Statistical
Method	ZEUS (Ours)	Chronos-2 (2025)	TimesFM2.5 (2023)	TiRex (2025)	Xihe _u (2025)	FlowState (2025)	Kairos _b (2025)	Moirai2 (2024)	Toto (2025)	Sundial (2025b)	PatchTST (2023)	DLinear (2023)	Seasonal Naive
MASE	0.693	<u>0.698</u>	0.705	0.716	0.701	0.726	0.742	0.728	0.750	0.750	0.849	1.061	1.000
CRPS	0.480	<u>0.485</u>	0.490	0.488	0.488	0.502	0.548	0.516	0.517	0.559	0.587	0.846	1.000

 Table 3. Imputation results under random and block masking, where {12.5%, 25%, 37.5%, 50%} points are masked. Reported metrics are averaged over four mask ratios. Best and second-best results are in **bold** and underlined, respectively. See Table D.3 for full results.

Models		Time Series Foundation Models (Zero-shot)								Task-Specific Models (Supervised)									
		ZEUS (Ours)		MOMENT (2024)		Timer (2024b)		UniTS (2024)		GPT4TS (2023)		ModernTCN (2024)		TimesNet (2023)		PatchTST (2023)		DLinear (2023)	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	Random	0.079	0.175	0.382	0.398	0.484	0.451	0.788	0.614	0.103	0.215	<u>0.086</u>	0.202	0.089	<u>0.198</u>	0.131	0.239	0.167	0.279
	Block	0.115	0.202	0.412	0.414	0.507	0.460	0.767	0.613	0.135	0.243	0.104	<u>0.222</u>	<u>0.111</u>	0.218	0.193	0.283	0.241	0.337
ETTh2	Random	0.056	0.136	0.166	0.276	0.182	0.283	0.589	0.537	0.065	0.171	<u>0.058</u>	0.162	0.059	<u>0.161</u>	0.069	0.169	0.147	0.261
	Block	0.067	0.151	0.192	0.294	0.192	0.290	0.618	0.549	0.082	0.194	<u>0.073</u>	<u>0.185</u>	0.078	<u>0.185</u>	0.093	0.198	0.278	0.353
ETTm1	Random	0.038	0.116	0.275	0.335	0.676	0.506	0.730	0.590	0.065	0.169	<u>0.051</u>	<u>0.152</u>	0.053	<u>0.152</u>	0.058	0.157	0.086	0.201
	Block	0.064	0.142	0.322	0.360	0.698	0.515	0.734	0.592	0.094	0.199	<u>0.074</u>	0.181	0.076	<u>0.178</u>	0.117	0.212	0.186	0.290
ETTm2	Random	0.027	0.084	0.103	0.214	0.125	0.239	0.505	0.498	0.034	0.119	<u>0.032</u>	<u>0.114</u>	<u>0.032</u>	<u>0.114</u>	0.035	0.115	0.102	0.214
	Block	0.035	0.099	0.125	0.233	0.131	0.245	0.537	0.513	0.048	0.145	0.046	0.141	<u>0.045</u>	<u>0.138</u>	0.052	0.143	0.191	0.292
ECL	Random	0.045	0.132	0.304	0.419	0.412	0.499	0.888	0.771	0.114	0.235	0.104	0.230	0.104	0.223	<u>0.090</u>	<u>0.212</u>	0.111	0.237
	Block	0.058	0.146	0.327	0.432	0.441	0.516	0.861	0.753	0.124	0.249	0.122	0.247	<u>0.110</u>	<u>0.229</u>	0.116	0.238	0.150	0.274
Weather	Random	0.030	0.035	0.083	0.142	0.112	0.168	0.207	0.288	0.036	0.072	<u>0.034</u>	<u>0.064</u>	<u>0.034</u>	0.067	0.036	<u>0.064</u>	0.102	0.214
	Block	0.036	0.042	0.093	0.154	0.119	0.175	0.211	0.291	0.044	0.084	0.046	0.085	<u>0.043</u>	<u>0.082</u>	0.051	0.085	0.093	0.169

ing task. We consider four different prediction horizons {96, 192, 336, 720}, where the best context length is chosen through hyperparameter search. Performance is evaluated using MSE and MAE.

Results As shown in Table 1, ZEUS demonstrates strong overall performance among advanced models. Compared with the previous state-of-the-art model, ZEUS achieves an averaged reduction of **9.0%** on MSE and **2.3%** on MAE. Moreover, ZEUS overcomes the common limitation where TSFMs lag behind task-specific models in zero-shot forecasting. When compared to the best-performing TSFM under the zero-shot settings, Timer (Liu et al., 2024b), ZEUS achieves substantial improvements, reducing MSE by **40.3%** and MAE by **25.3%**.

4.2. Probabilistic Forecasting

Setups Beyond point forecasting, we evaluate our model on the GIFT-Eval benchmark (Aksu et al., 2024) for probabilistic forecasting, which comprises 97 prediction tasks spanning short-, medium-, and long-term forecasting, providing a comprehensive assessment of predictive performance. Following the official protocol, we report MASE and CRPS as evaluation metrics.

Results Table 2 reports the aggregated performance across 97 tasks on GIFT-Eval. Among the advanced pretrained

forecasting models, including Chronos-2 (Ansari et al., 2025), TimesFM2.5 (Das et al., 2023), and TiRex (Auer et al., 2025), ZEUS ranks first on both MASE and CRPS, highlighting its strong zero-shot forecasting capabilities.

4.3. Imputation

Setups Time series imputation aims to fill in missing values of time series and is critical in real-world applications. Following previous work (Liu et al., 2024b; Zhang et al., 2025), we evaluate ZEUS on the ETT, ECL and Weather datasets by masking {12.5%, 25%, 37.5%, 50%} time steps in sequences of length 192. Similar to point forecasting tasks, We use MSE and MAE as evaluation metrics.

Existing TSFMs (Liu et al., 2024b; Goswami et al., 2024; Zhang et al., 2025) are typically evaluated under patch-wise missing patterns. However, real-world time series exhibit more diverse missing behaviors, including both point-wise missing and contiguous segment missing with variable lengths (Yu et al., 2025b). To better reflect practical scenarios, we adopt two masking strategies: (1) random masking, which has been widely adopted in prior task-specific models (Wu et al., 2023; donghao & xue, 2024), and (2) block masking, where the lengths of contiguous missing segments are sampled from a Geometry distribution with $p = 0.125$.

Results As shown in Table 3, ZEUS consistently achieves superior performance across all datasets and in zero-shot,

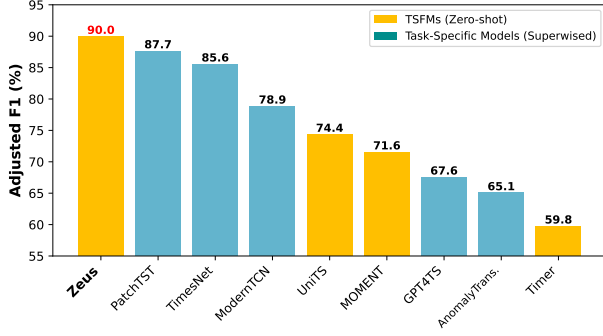


Figure 4. Averaged adjusted F1 score on 42 UCR anomaly detection datasets. See Table 9 for full results.

demonstrating strong robustness to variable-length missing patterns. Patch-based TSFMs (MOMENT, UniTS, and Timer) perform substantially worse than task-specific models, mainly due to the mismatch between their patch-wise pretraining objective and point-level missing patterns at inference, which results in an out-of-distribution scenario. In contrast, ZEUS effectively handles both random and block missing scenarios. Even compared with the strongest task-specific models, ZEUS reduces the averaged MSE by **24.4%** under random masking and **18.8%** under block masking.

4.4. Anomaly Detection

Setups To provide a comprehensive evaluation, we conduct experiments on a collection of 42 datasets sourced from the UCR anomaly archive (Wu & Keogh, 2021). This subset that has been adopted in previous studies (Goswami et al., 2024), covering a sufficiently broad range of domains and data sources. We partition the time series into non-overlapping detection windows (Xu et al., 2022) and use the reconstruction error as the criterion for anomaly detection (Wu et al., 2023). The window size is selected via hyperparameter search and the results of all baselines are reported under the optimal settings. We use adjusted F1-score as the evaluation metric (Wu et al., 2023).

Results Figure 4 presents the overall performance of anomaly detection tasks. ZEUS ranks first among all baselines across 42 UCR datasets. Under the zero-shot setting, ZEUS outperforms advanced task-specific models trained in full-shot. Moreover, ZEUS achieves a **21.0%** improvement in F1-score over UniTS (Gao et al., 2024), the second-best performing TSFM on this benchmark.

4.5. Classification

Setups We conduct evaluations on 26 UEA classification datasets (Bagnall et al., 2018). Prior TSFMs relies on fine-tuning (Gao et al., 2024; Zhang et al., 2025) or linear

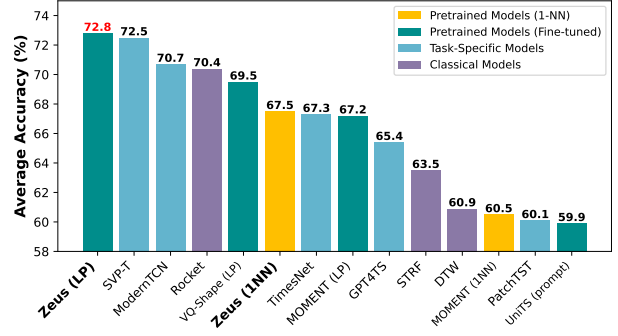


Figure 5. Averaged accuracy on 26 UEA classification datasets, where *LP* denotes linear probing and *prompt* denotes fine-tuning on prompt tokens. See Table 10 for full results.

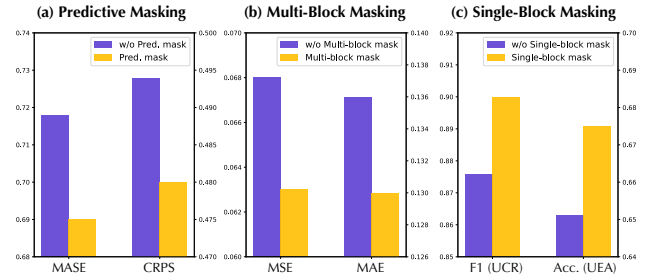


Figure 6. Ablation results of ZEUS.

probing (Goswami et al., 2024) for classification tasks. In our evaluation, ZEUS is primarily assessed in a tuning-free setting using a non-parametric 1-nearest neighbor (1-NN) classifier for evaluation, with optional PCA whitening applied for feature normalization. In addition, we also report linear probing results to assess the linear separability of the learned representations and to facilitate comparison with existing methods. Following prior work (Goswami et al., 2024), we use the accuracy as the evaluation metric.

Results As shown in Figure 5, ZEUS achieves superior performance across the 26 UEA classification datasets under different evaluation protocols. In particular, ZEUS with linear probing attains the highest averaged accuracy among all compared methods, outperforming both task-specific models and other pretrained baselines. Notably, ZEUS with 1-NN evaluation also delivers competitive performance, being on par with advanced models such as TimesNet and MOMENT-LP, while outperforming MOMENT under the same 1-NN setting by a clear margin of 7.0 percentage points. Overall, these results highlight the effectiveness and generalizability of ZEUS’s pretrained representations.

4.6. Ablation Study

In this section, we investigate the effectiveness of our proposed MOTM by ablating individual masking strategies.

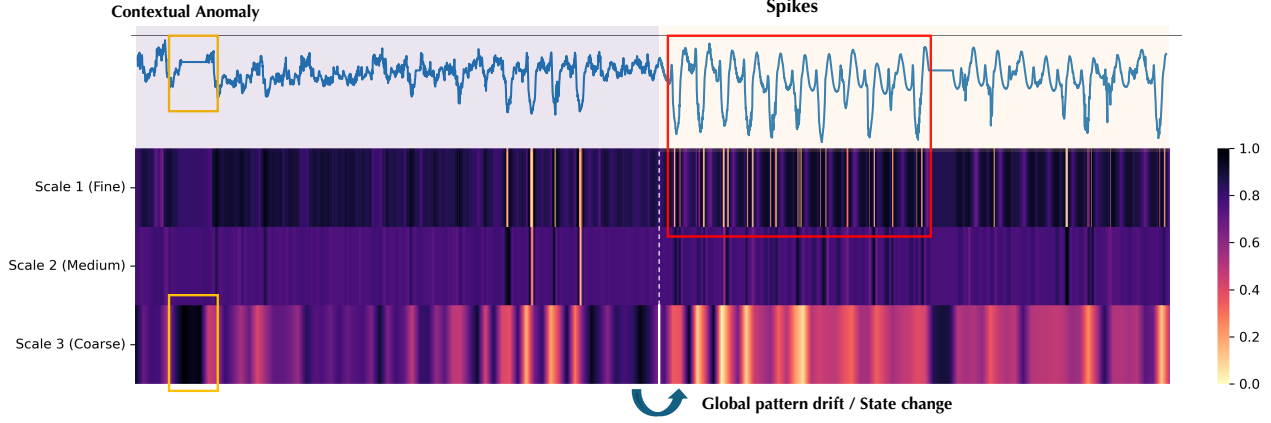


Figure 7. Multi-scale feature-norm heatmaps, which illustrates the roles of different scales: fine-scale representations are sensitive to local variations and extreme values (red boxes), mid-scale stripe patterns capture intrinsic periodicity, and coarse-scale representations model global pattern shifts (white vertical line) and contextual anomalies (yellow boxes).

When a specific mask is removed, we keep the expected masking ratio fixed by proportionally reallocating it to the other masks. As illustrated in Figure 6 (a), removing the predictive mask leads to a clear performance drop on the GiftEval benchmark, validating its crucial role in enhancing extrapolation capabilities and forecasting precision. Similarly, Figure 6 (b) shows the absence of multi-block mask degrades ZEUS’s performance on imputation tasks, indicating its importance for interpolation and local pattern reconstruction. Furthermore, the removal of the single-block mask leads to consistent performance degradation in both anomaly detection and classification, as shown in Figure 6 (c), suggesting that single-block reconstruction guides the model to capture global consistency.

4.7. Model Analysis

Representation Analysis We conduct a representation analysis to illustrate the roles of different temporal scales in ZEUS. Specifically, we select a sample sequence from the ETTm1 dataset and visualize the feature-norm heatmaps of the representations at multiple scales, as shown in Figure 7.

It can be observed that fine-scale representations are highly sensitive to local variations and extreme values. The regions highlighted by red boxes indicate that these representations effectively capture each negative spike in the sequence. In contrast, mid-scale representations are less responsive to local fluctuations; instead, their stripe-like patterns reflect the underlying periodic structures. Coarse-scale representations clearly characterize global pattern changes in the sequence. For instance, the white vertical line marks the transition from small-amplitude fluctuations to periodic negative impulses, which is reflected by a color change in the heatmap from purple to yellow. Moreover, the regions highlighted by

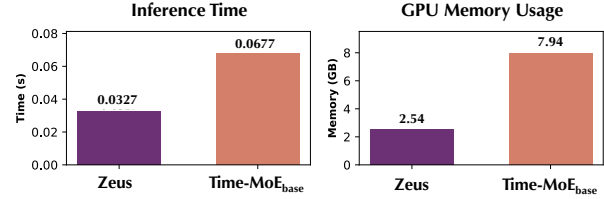


Figure 8. Efficiency comparison between ZEUS and Time-MoE_{base}, two point-tokenized models with comparable model size. Results are averaged over 1,000 runs on sequences of length $L=4096$.

yellow boxes demonstrate that large-scale representations are effective in capturing contextual anomalies.

Efficiency Analysis Conventional point-tokenized Transformers suffers from the high computational cost, as processing a sequence of length L with dimension d over N layers incurs $\mathcal{O}(NL^2d)$ computations due to full-resolution self-attention. In contrast, ZEUS performs most attention computations at coarser temporal resolutions, resulting in a reduced complexity of

$$\mathcal{C}_{\text{ZEUS}} = \sum_i \mathcal{O}\left(N_i \left(\frac{L}{s_i}\right)^2 d_i\right), \quad (6)$$

where N_i and d_i denote the number of layers and the hidden dimension at scale s_i . Under our model configuration (Appendix A), ZEUS achieves a $3.8\times$ reduction in self-attention FLOPs compared to a vanilla Transformer with the same depth (Appendix C.2). Additionally, we conduct an empirical efficiency comparison between ZEUS and Time-MoE_{base} (Shi et al., 2025), a point-tokenized pretrained model with approximately 113M parameters. Both models are evaluated

under the same environment with FlashAttention enabled, ensuring a fair comparison. As shown in Figure 8, ZEUS is $2.1\times$ faster and $3.1\times$ memory efficient than Time-MoE_{base}, demonstrating its scalability over long context.

5. Conclusion

In this work, we propose ZEUS, a unified multi-scale Transformer with point-wise tokenization and a U-shaped hierarchy that effectively balances fine-grained temporal fidelity with long-sequence efficiency. Complementing the architectural design, we introduce MOTM, a multi-objective temporal masking strategy that jointly supports extrapolation, interpolation, and global abstraction within a single pretraining framework. Moreover, we construct Aegis-Syn, a synthetic dataset that extends KernelSynth with richer temporal patterns. Extensive experiments across five representative downstream tasks demonstrate that ZEUS achieves strong and consistent performance in a fully tuning-free setting, highlighting its potential as a truly general-purpose TSFM. We believe this work takes a meaningful step toward versatile and scalable foundation models for time series analysis. Future work will explore multivariate modeling and extend the proposed framework to a broader range of downstream tasks. See Appendix F for detailed discussion on limitations and future directions.

Acknowledgments

This work is supported by the NSFC under Grant Nos.62372430 and 62502505, the Youth Innovation Promotion Association CAS No.2023112, the Postdoctoral Fellowship Program of CPSF under Grant Number GZC20251078, the China Postdoctoral Science Foundation No.2025M77154 and HUA Innovation fundings. We sincerely thank all the anonymous reviewers who generously contributed their time and efforts.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Abeywickrama, S., Eldele, E., Wu, M., Li, X., and Yuen, C. Entropy guided dynamic patch segmentation for time series transformers, 2026. URL <https://arxiv.org/abs/2509.26157>.

Aksu, T., Woo, G., Liu, J., Liu, X., Liu, C., Savarese, S., Xiong, C., and Sahoo, D. Gift-eval: A benchmark for gen-

eral time series forecasting model evaluation. In *NeurIPS Workshop on Time Series in the Age of Large Models*, 2024.

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., et al. Chronos: Learning the language of time series. *arXiv preprint arXiv:2403.07815*, 2024.

Ansari, A. F., Shchur, O., Kuken, J., Auer, A., Han, B., Mercado, P., Rangapuram, S. S., Shen, H., Stella, L., Zhang, X., Goswami, M., Kapoor, S., Maddix, D. C., Gueron, P., Hu, T., Yin, J., Erickson, N., Desai, P. M., Wang, H., Rangwala, H., Karypis, G., Wang, Y., and Bohlke-Schneider, M. Chronos-2: From Univariate to Universal Forecasting. *arXiv.org*, abs/2510.15821, 2025.

Auer, A., Podest, P., Klotz, D., Böck, S., Klambauer, G., and Hochreiter, S. Tirex: Zero-Shot Forecasting Across Long and Short Horizons with Enhanced In-Context Learning. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, volume abs/2505.23719, 2025.

Bagnall, A., Dau, H. A., Lines, J., Flynn, M., Large, J., Bostrom, A., Southam, P., and Keogh, E. The uea multivariate time series classification archive, 2018, 2018. URL <https://arxiv.org/abs/1811.00075>.

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., and Joulin, A. Unsupervised learning of visual features by contrasting cluster assignments. *Advances in neural information processing systems*, 33:9912–9924, 2020.

Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pp. 1597–1607. PMLR, 2020.

Chen, Y., Hu, B., Keogh, E., and Batista, G. E. Dtw-d: time series semi-supervised learning from a single example. In *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*, pp. 383–391, 2013.

Cohen, B., Khwaja, E., Doubli, Y., Lemaachi, S., Lettieri, C., Masson, C., Miccinilli, H., Ramé, E., Ren, Q., Ros-tamizadeh, A., Terrail, J. O. d., Toon, A.-M., Wang, K., Xie, S., Xu, Z., Zhukova, V., Asker, D., Talwalkar, A., and Abou-Amal, O. This Time is Different: An Observability Perspective on Time Series Foundation Models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, volume abs/2505.14766, 2025.

Dao, T. Flashattention-2: Faster Attention with Better Parallelism and Work Partitioning. In *International Conference on Learning Representations*, 2024.

- Das, A., Kong, W., Sen, R., and Zhou, Y. A decoder-only foundation model for time-series forecasting. *arXiv preprint arXiv:2310.10688*, 2023.
- Dempster, A., Petitjean, F., and Webb, G. I. Rocket: exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery*, 34(5):1454–1495, 2020.
- Deng, J., Chen, X., Jiang, R., Song, X., and Tsang, I. W. A multi-view multi-task learning framework for multivariate time series forecasting. *IEEE Transactions on Knowledge and Data Engineering*, 35(8):7665–7680, 2022.
- Deng, J., Chen, X., Jiang, R., Yin, D., Yang, Y., Song, X., and Tsang, I. W. Disentangling structured components: Towards adaptive, interpretable and scalable time series forecasting. *IEEE Transactions on Knowledge and Data Engineering*, 36(8):3783–3800, 2024.
- donghao, L. and xue, w. Moderntcn: A Modern Pure Convolution Structure for General Time Series Analysis. In *The Twelfth International Conference on Learning Representations*, 2024.
- Ekambaram, V., Jati, A., Dayama, P., Mukherjee, S., Nguyen, N. H., Gifford, W. M., Reddy, C., and Kalagnanam, J. Tiny Time Mixers (TTMs): Fast Pre-trained Models for Enhanced Zero/Few-Shot Forecasting of Multivariate Time Series. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- Eldele, E., Ragab, M., Chen, Z., Wu, M., and Li, X. Tslanet: Rethinking transformers for time series representation learning. In *International Conference on Machine Learning*, pp. 12409–12428. PMLR, 2024.
- Feng, K., Lan, S., Fang, Y., He, W., Ma, L., Lu, X., and Ren, K. Kairos: Towards Adaptive and Generalizable Time Series Foundation Models. *arXiv.org*, abs/2509.25826, 2025.
- Fu, Y., Wang, F., Shao, Z., Diao, B., Wu, L., An, Z., Yu, C., Li, Y., and Xu, Y. On the Integration of Spatial-Temporal Knowledge: A Lightweight Approach to Atmospheric Time Series Forecasting. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- Fu, Y., Shao, Z., Yu, C., Li, Y., An, Z., Wang, Q., Xu, Y., and Wang, F. Selective learning for deep time series forecasting. *Advances in neural information processing systems*, 38:98084–98115, 2026.
- Gao, S., Koker, T., Queen, O., Hartvigsen, T., Tsiligkaridis, T., and Zitnik, M. Units: A Unified Multi-Task Time Series Model. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- Gneiting, T. and Raftery, A. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102:359–378, 03 2007. doi: 10.1198/016214506000001437.
- Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., and Dubrawski, A. Moment: A family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*, 2024.
- Graf, L., Ortner, T., WoŁsniak, S., Pantazi, A., et al. Flow-state: Sampling rate invariant time series forecasting. *arXiv preprint arXiv:2508.05287*, 2025.
- Gruver, N., Finzi, M., Qiu, S., and Wilson, A. G. Large language models are zero-shot time series forecasters. *Advances in Neural Information Processing Systems*, 36, 2023.
- Hills, J., Lines, J., Baranauskas, E., Mapp, J., and Bagnall, A. Classification of time series by shapelet transformation. *Data mining and knowledge discovery*, 28(4):851–881, 2014.
- Huang, J., Xu, Y., Wang, Q., Wang, Q. C., Liang, X., Wang, F., Zhang, Z., Wei, W., Zhang, B., Huang, L., et al. Foundation models and intelligent decision-making: Progress, challenges, and perspectives. *The Innovation*, 2025.
- Huang, Q., Shen, L., Zhang, R., Cheng, J., Ding, S., Zhou, Z., and Wang, Y. Hdmixer: hierarchical dependency with extendable patch for multivariate time series forecasting. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence*, AAAI’24/IAAI’24/EAAI’24. AAAI Press, 2024. ISBN 978-1-57735-887-9. doi: 10.1609/aaai.v38i11.29155. URL <https://doi.org/10.1609/aaai.v38i11.29155>.
- Hyndman, R. J. and Koehler, A. B. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22, 2006.
- Jin, M., Wang, S., Ma, L., Chu, Z., Zhang, J. Y., Shi, X., Chen, P.-Y., Liang, Y., Li, Y.-F., Pan, S., and Wen, Q. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models. In *The Twelfth International Conference on Learning Representations*, 2024.

- Joshi, M., Chen, D., Liu, Y., Weld, D. S., Zettlemoyer, L., and Levy, O. Spanbert: Improving Pre-training by Representing and Predicting Spans. *Transactions of the Association for Computational Linguistics*, 8:64–77, 2020.
- Kim, T., Kim, J., Tae, Y., Park, C., Choi, J.-H., and Choo, J. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations*, 2021.
- Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Le, X.-M., Tran, M.-T., and Huynh, V.-N. Learning perceptual position-aware shapelets for time series classification. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pp. 53–69. Springer, 2022.
- Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V., and Zettlemoyer, L. Bart: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 7871–7880, 2020.
- Li, J., Zhou, P., Xiong, C., and Hoi, S. C. H. Prototypical Contrastive Learning of Unsupervised Representations. In *International Conference on Learning Representations*, 2021.
- Liu, X., Li, X., Li, Y., Tang, F., and Zhao, M. Rtdetector: Deep transformer networks for time series anomaly detection based on reconstruction trend. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25, J. Kwok, Ed. International Joint Conferences on Artificial Intelligence Organization*, volume 8, pp. 5788–5796, 2025a.
- Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., and Long, M. itransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *The Twelfth International Conference on Learning Representations*, 2024a.
- Liu, Y., Zhang, H., Li, C., Huang, X., Wang, J., and Long, M. Timer: Generative pre-trained transformers are large time series models. In *Forty-first International Conference on Machine Learning*, 2024b.
- Liu, Y., Zhang, K., Li, Y., Yan, Z., Gao, C., Chen, R., Yuan, Z., Huang, Y., Sun, H., Gao, J., et al. Sora: A review on background, technology, limitations, and opportunities of large vision models. *arXiv preprint arXiv:2402.17177*, 2024c.
- Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J., and Long, M. Sundial: A Family of Highly Capable Time Series Foundation Models. In *Forty-second International Conference on Machine Learning*, 2025b.
- Luo, Y., Cai, X., Zhang, Y., Xu, J., et al. Multivariate time series imputation with generative adversarial networks. *Advances in neural information processing systems*, 31, 2018.
- Masserano, L., Ansari, A. F., Han, B., Zhang, X., Faloutsos, C., Mahoney, M. W., Wilson, A. G., Park, Y., Ranganapuram, S. S., Maddix, D. C., and Wang, B. Enhancing foundation models for time series forecasting via wavelet-based tokenization. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=B6WalMoQJW>.
- Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*, 2023.
- Niu, Y., Deng, J., and Tong, Y. Phaseformer: From patches to phases for efficient and effective time series forecasting. *arXiv preprint arXiv:2510.04134*, 2025.
- OpenAI, R. Gpt-4 technical report. arxiv 2303.08774. *View in Article*, 2:13, 2023.
- Park, Y., Maddix, D. C., Aubet, F.-X., Kan, K. K., Gasthaus, J., and Wang, Y. Learning quantile functions without quantile crossing for distribution-free time series forecasting. In *International Conference on Artificial Intelligence and Statistics*, 2021. URL <https://api.semanticscholar.org/CorpusID:244103049>.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019.
- Qin, G., Chen, Z., Liu, Y., Shi, Z., Liu, H., Huang, X., Wang, J., and Long, M. Cora: Covariate-aware adaptation of time series foundation models, 2025. URL <https://arxiv.org/abs/2510.12681>.
- Ramesh, A., Pavlov, M., Goh, G., Gray, S., Voss, C., Radford, A., Chen, M., and Sutskever, I. Zero-shot text-to-image generation. In *International conference on machine learning*, pp. 8821–8831. Pmlr, 2021.
- Šabić, E., Keeley, D., Henderson, B., and Nannemann, S. Healthcare and anomaly detection: using machine learning to predict anomalies in heart rate data. *Ai & Society*, 36(1):149–158, 2021.

- Shao, Z., Zhang, Z., Wang, F., Wei, W., and Xu, Y. Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management, CIKM '22*, pp. 4454–4458. Association for Computing Machinery, 2022. ISBN 9781450392365.
- Shao, Z., Wang, F., Xu, Y., Wei, W., Yu, C., Zhang, Z., Yao, D., Sun, T., Jin, G., Cao, X., et al. Exploring progress in multivariate time series forecasting: Comprehensive benchmarking and heterogeneity analysis. *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- Shao, Z., Li, Y., Wang, F., Yu, C., Fu, Y., Qian, T., Xu, B., Diao, B., Xu, Y., and Cheng, X. Blast: Balanced Sampling Time Series Corpus for Universal Forecasting Models. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, pp. 2502–2513. ACM, 2025a.
- Shao, Z., Wang, F., Sun, T., Yu, C., Fang, Y., Jin, G., An, Z., Liu, Y., Qu, X., and Xu, Y. Hutformer: Hierarchical u-net transformer for long-term traffic forecasting. *Communications in Transportation Research*, 5:100218, 2025b.
- Shazeer, N. Glu variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020.
- Shi, X., Wang, S., Nie, Y., Li, D., Ye, Z., Wen, Q., and Jin, M. Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., and Liu, Y. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Sun, Y., Fang, Y., Zhu, Z., Li, J., Liu, Y., Deng, Q., Zhou, J., Yu, H., Lu, X., and Ma, L. Xihe: Scalable zero-shot time series learner via hierarchical interleaved block attention. *arXiv preprint arXiv:2510.21795*, 2025.
- Tan, M., Merrill, M. A., Gupta, V., Althoff, T., and Hartvigsen, T. Are language models actually useful for time series forecasting? In *Advances in Neural Information Processing Systems*, volume 37, pp. 60162–60191, 2024.
- Tang, P. and Zhang, W. Unlocking the power of patch: Patch-based mlp for long-term time series forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(12):12640–12648, Apr. 2025.
- van der Maaten, L. and Hinton, G. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(86):2579–2605, 2008. URL <http://jmlr.org/papers/v9/vandermaaten08a.html>.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Vrigkas, M., Nikou, C., and Kakadiaris, I. A. A review of human activity recognition methods. *Frontiers in Robotics and AI*, 2:28, 2015.
- Wang, F., Li, Y., Shao, Z., Yu, C., Fu, Y., An, Z., Xu, Y., and Cheng, X. Aries: Relation assessment and model recommendation for deep time series forecasting. *arXiv preprint arXiv:2509.06060*, 2025.
- Wen, Y., Ma, T., Weng, L., Nguyen, L., and Julius, A. A. Abstracted shapes as tokens-a generalizable and interpretable model for time-series classification. *Advances in Neural Information Processing Systems*, 37:92246–92272, 2024.
- Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., and Sahoo, D. Unified training of universal time series forecasting transformers. *arXiv preprint arXiv:2402.02592*, 2024.
- Wu, H., Xu, J., Wang, J., and Long, M. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Advances in Neural Information Processing Systems*, 34:22419–22430, 2021.
- Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., and Long, M. Timesnet: Temporal 2d-Variation Modeling for General Time Series Analysis. In *International Conference on Learning Representations*, 2023.
- Wu, R. and Keogh, E. Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress. *IEEE Transactions on Knowledge and Data Engineering*, 2021.
- Xiong, R., Yang, Y., He, D., Zheng, K., Zheng, S., Xing, C., Zhang, H., Lan, Y., Wang, L., and Liu, T. On layer normalization in the transformer architecture. In *International Conference on Machine Learning*, pp. 10524–10533. PMLR, 2020.
- Xu, J., Wu, H., Wang, J., and Long, M. Anomaly transformer: Time series anomaly detection with association discrepancy. In *International Conference on Learning Representations*, 2022.
- Xu, Y., Liu, X., Cao, X., Huang, C., Liu, E., Qian, S., Liu, X., Wu, Y., Dong, F., Qiu, C.-W., et al. Artificial

- intelligence: A powerful paradigm for scientific research. *The Innovation*, 2(4), 2021.
- Yi, K., Zhang, Q., Fan, W., Wang, S., Wang, P., He, H., An, N., Lian, D., Cao, L., and Niu, Z. Frequency-domain mlps are more effective learners in time series forecasting. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S. (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 76656–76679. Curran Associates, Inc., 2023.
- Yu, C., Wang, F., Shao, Z., Sun, T., Wu, L., and Xu, Y. Dsformer: A double sampling transformer for multivariate time series long-term prediction. *CIKM '23*, pp. 3062–3072. Association for Computing Machinery, 2023. ISBN 9798400701245. doi: 10.1145/3583780.3614851.
- Yu, C., Wang, F., Wang, Y., Shao, Z., Sun, T., Yao, D., and Xu, Y. Mgsformer: A multi-granularity spatiotemporal fusion transformer for air quality prediction. *Information Fusion*, 113:102607, 2025a. ISSN 1566-2535. doi: <https://doi.org/10.1016/j.inffus.2024.102607>. URL <https://www.sciencedirect.com/science/article/pii/S1566253524003853>.
- Yu, C., Wang, F., Yang, C., Shao, Z., Sun, T., Qian, T., Wei, W., An, Z., and Xu, Y. Merlin: Multi-View Representation Learning for Robust Multivariate Time Series Forecasting with Unfixed Missing Rates. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, pp. 3633–3644. ACM, 2025b.
- Yue, Z., Wang, Y., Duan, J., Yang, T., Huang, C., Tong, Y., and Xu, B. Ts2vec: Towards universal representation of time series. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 8980–8987, 2022.
- Zeng, A., Chen, M., Zhang, L., and Xu, Q. Are transformers effective for time series forecasting? In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pp. 11121–11128, 2023.
- Zhang, B. and Sennrich, R. Root Mean Square Layer Normalization. In *Conference on Neural Information Processing Systems*, pp. 12360–12371, 2019.
- Zhang, H., Liu, Y., Qiu, Y., Liu, H., Pei, Z., Wang, J., and Long, M. Timesbert: A bert-style foundation model for time series understanding, 2025. URL <https://arxiv.org/abs/2502.21245>.
- Zhang, Y. and Yan, J. Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In *The eleventh international conference on learning representations*, 2023.
- Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., and Zhang, W. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pp. 11106–11115, 2021.
- Zhou, T., Niu, P., Wang, X., Sun, L., and Jin, R. One fits all: Power general time series analysis by pretrained lm. *arXiv preprint arXiv:2302.11939*, 2023.
- Zuo, R., Li, G., Choi, B., Bhowmick, S. S., Mah, D. N.-y., and Wong, G. L. Svp-t: A shape-level variable-position transformer for multivariate time series classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 11497–11505, 2023.
- Özgün Turgut, Müller, P., Menten, M. J., and Rueckert, D. Towards generalisable time series understanding across domains, 2025. URL <https://arxiv.org/abs/2410.07299>.

A. Implementation Details

A.1. Model Configuration

ZEUS has approximately 100M parameters, and the detailed model configurations are summarized in Table 4. We set the maximum context length to 4096. By using FlashAttention (Dao, 2024) and preventing padded tokens from participating in attention, we ensure that the additional padding does not incur significant computational overhead. The model are trained for 200,000 steps with a global batch size of 512. We employ the AdamW optimizer (Kingma & Ba, 2014) with a cosine learning rate schedule and a warmup phase of 10,000 steps. The initial learning rate is set to 1×10^{-3} . All experiments are implemented using PyTorch (Paszke et al., 2019) and trained on 4 NVIDIA H100 GPUs.

Table 4. Hyperparameters of ZEUS.

Hyperparameter	Value
Scales	[1, 8, 32, 8, 1]
# Layers	[1, 3, 3, 3, 2]
Hidden size	[384, 768, 768, 768, 384]
# Heads	[6, 12, 12, 12, 6]
Intermediate size	[1536, 3072, 3072, 3072, 1536]
# Parameters	100M

A.2. Unified Downstream Task Formulation

This section describes how ZEUS is applied to various downstream tasks.

Forecasting Given a historical time series context $\mathbf{x} \in \mathbb{R}^{T \times C}$, where T denotes the context length and C the number of variables, ZEUS performs forecasting by formulating the task as a masked sequence completion problem. ZEUS adopt channel independent strategy that treat $\mathbf{x}$ as C univariate time series.

To perform forecasting on a horizon H , the context is concatenated with H [MASK] tokens, i.e., $\tilde{\mathbf{x}} = [\mathbf{x}, \mathbf{M}] \in \mathbb{R}^{(T+H) \times C}$, and then fed to ZEUS. The output of ZEUS is a quantile reconstruction tensor $\mathbf{z} \in \mathbb{R}^{(T+H) \times C \times |Q|}$, where $|Q|$ denotes the number of quantile levels. We retain the predictions at the final H steps as the probabilistic forecast $\mathbf{y} \in \mathbb{R}^{H \times C \times |Q|}$, and the point forecast is obtained by averaging predictions across all quantiles.

Imputation For the imputation task, ZEUS follows the same reconstruction paradigm as used during pretraining. Missing values are replaced with [MASK] tokens and fed into ZEUS, which produces quantile predictions for all time steps. The outputs at masked positions are used as imputed values, with point estimates obtained by averaging across quantiles.

Anomaly Detection ZEUS supports anomaly detection under two complementary paradigms: *prediction-based* and *reconstruction-based*. The prediction-based approach detects anomalies by forecasting future values and measuring prediction errors, which is well suited for streaming scenarios where only past observations are available. In contrast, the reconstruction-based approach masks a target segment and reconstructs it using both past and future context, enabling more accurate detection by exploiting full contextual information. In this work, we provide a formal description of the reconstruction-based formulation, while the prediction-based variant follows an analogous procedure.

Given a historical time series $\mathbf{X}_{1:L} \in \mathbb{R}^{L \times C}$ and a target window size $T < L$, we partition the sequence into non-overlapping subsequences using a sliding window with stride T . For each target segment $\mathbf{X}_{t:t+T}$, we construct the input by masking the target window and concatenating it with its surrounding context, forming $\tilde{\mathbf{X}} = [\mathbf{X}_{t-W:t}, \mathbf{M}, \mathbf{X}_{t+T:t+T+W}]$, where W denotes the context length and $\mathbf{M}$ denotes T consecutive [MASK] tokens. The resulting sequence is then fed into ZEUS to reconstruct the masked segment.

The model outputs quantile predictions $\mathbf{z} \in \mathbb{R}^{T \times C \times |Q|}$ for the target window. An anomaly score is computed based on the reconstruction error, with higher errors indicating a higher likelihood of anomalies. By default, we use MAE as the reconstruction error for anomaly scoring. However, for sequences containing impulsive patterns, we selectively adopt relative MAE as the error metric, which normalizes the absolute error by the signal magnitude. This choice prevents large but expected impulses from being erroneously identified as anomalies.

Classification ZEUS performs classification using a 1-nearest neighbor (1-NN) classifier on the learned representations. Specifically, given the train dataset $\mathcal{D}_{train}$ and a sample, ZEUS predict label $\hat{y}$ with its hidden states $\mathbf{h} \in \mathbb{R}^{T \times C \times d}$ by

$$\mathbf{z} = \text{Flatten}(\text{GlobalPool}(\mathbf{h})), \quad (7)$$

$$i^* = \arg \max_{i \in \mathcal{D}_{train}} \text{sim}(\mathbf{z}, \mathbf{z}_i), \quad (8)$$

$$\hat{y} = y_{i^*}, \quad (9)$$

where $\text{GlobalPool}(\cdot)$ aggregates the temporal dimension, for which we default to max pooling, $\text{Flatten}(\cdot)$ flattens the channel dimension, and $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity. $\mathbf{h}$ can be taken from a specific scale or formed by concatenating representations from multiple scales. In practice, we typically use the representation from the penultimate scale ($s_4 = 8$) or the coarsest scale ($s_3 = 32$).

As an alternative to 1-NN classification, we also consider a linear probe applied on top of $\mathbf{z}$, where the backbone of ZEUS is frozen during training.

A.3. Technical Details of MOTM

This section provides the technical details of MOTM that is used to pretrain ZEUS.

Masking Ratio The overall masking ratio is sampled from a uniform distribution, i.e., $p \sim \mathcal{U}(0, 0.5)$, with the expected rate being 0.25. A low masking ratio is used to simulate short-term forecasting scenarios, where the model need to accurately infer future dynamics from long historical context. In contrast, a high masking ratio poses a more challenging setting and is used to emulate imputation scenarios with severe missingness. Compared to a fixed masking ratio that is typically adopted by prior work (Goswami et al., 2024; Zhang et al., 2025), a variable masking ratio prevents the model from overfitting to a single missingness level and improves robustness across diverse missing rates.

Temporal Range To increase the diversity of contexts observed by the model, we sample variable sequence lengths and randomly crop the input sequences, padding them to a maximum context length of 4096. Specifically, the sequence length is sampled from a piecewise uniform distribution: with probability 0.2 from $[64, 512]$, 0.2 from $[513, 2048]$, and 0.6 from $[2049, 4096]$. This design ensures that short- and mid-range temporal dependencies are sufficiently observed, while most sequences fall into the long-range regime and require minimal padding, thereby balancing training efficiency.

B. Related Work

B.1. Systematic Comparison with Closely Related Work

In this section, we present a systematic comparison with closely related TSFMs. In this work, we use the term *time series foundation models* to denote pretrained models that are capable of supporting a wide range of downstream time series analysis tasks, in contrast to *pretrained forecasting models* that are designed exclusively for forecasting. In addition, some LLM-based approaches, such as GPT4TS (Zhou et al., 2023), do not involve pretraining on time series data. Existing studies further indicate that the performance gains do not primarily stem from the LLM backbone (Tan et al., 2024). Therefore, we classify these methods as task-specific models. While state-of-the-art pretrained forecasting models are included in the forecasting evaluations, this section focuses on comparisons among TSFMs.

Table 5 summarizes the key differences between ZEUS and representative TSFMs, including TimesBERT (Zhang et al., 2025), MOMENT (Goswami et al., 2024), UniTS (Gao et al., 2024), and Timer (Liu et al., 2024b), in terms of model architecture, pretraining scale, tokenization strategy, and downstream task support. Existing TSFMs predominantly adopt patch-based tokenization, whereas ZEUS employs point-wise tokenization to preserve fine-grained details. In addition, ZEUS is pretrained at a larger scale and with a substantially longer context length than prior TSFMs. As TimesBERT is not publicly available, it is not included as a baseline in our experiments.

Beyond these differences, a more salient distinction lies in downstream task coverage. ZEUS supports a substantially broader set of time series analysis tasks than existing TSFMs, enabling forecasting, imputation, anomaly detection, and classification in a tuning-free manner. In contrast, existing TSFMs typically support only a subset of downstream tasks

Table 5. Comparison with existing TSFMs. A dash (–) indicates that the information is unspecified or not explicitly reported. For downstream task support, ✓ denotes zero-shot capability, ✗ indicates unsupported task, and ○ indicates that fine-tuning is required.

	ZEUS (Ours)	TimesBERT (2025)	MOMENT (2024)	UniTS (2024)	Timer (2024b)
Architecture	Encoder-only	Encoder-only	Encoder-only	Encoder-only	Decoder-only
Model Size	100M	86M	40M, 125M, 385M	8M	29M, 50M, 67M
Pretraining Scale	300B	260B	1B	35M	28B
Context Length	4096	–	512	–	1440
Tokenization	Point	Patch	Patch	Patch	Patch
Open-sourced	✓	✗	✓	✓	✓
Supported Downstream Tasks					
Long-term Forecasting	✓	✗	○	○	✓
Short-term Forecasting	✓	○	✓	○	✓
Probabilistic Forecasting	✓	✗	✗	✗	✗
Imputation	✓	○	✓	○	○
Anomaly Detection	✓	○	✓	○	○
Classification	✓	○	○	○	✗

or rely on task-specific fine-tuning or adaptations. Overall, this comparison positions ZEUS as a more general-purpose foundation model with comprehensive and unified multi-task capabilities.

B.2. Multi-Scale Architectures

Multi-scale modeling has been widely explored in task-specific models (Zhang & Yan, 2023; Yu et al., 2023; 2025a; Deng et al., 2024; Shao et al., 2025b), where hierarchical representations are adopted to capture temporal dependencies at different resolutions. These studies have demonstrated the effectiveness of multi-scale designs for improving model performance. However, despite their success in task-specific settings, most existing TSFMs still rely on single-scale architectures. Only a few exceptions explored multi-scale structures, and ZEUS differs from these works fundamentally in both information flow and architectural design. For instance, TTM (Ekambaram et al., 2024) and Xihe (Sun et al., 2025) follows a fine-to-coarse paradigm, progressively aggregating local patterns, while Kairos (Feng et al., 2025) adopts a coarse-to-fine mixture-of-size patching strategy. In contrast, ZEUS is the first to introduce a fine-to-coarse-to-fine multi-scale architecture, which preserves point-level details while simultaneously enhancing long-sequence scalability.

B.3. Time Series Tokenization

Time series tokenization has been extensively studied in recent years. Beyond the widely adopted point tokens (Zhou et al., 2021; Deng et al., 2022) and patch tokens (Nie et al., 2023; Tang & Zhang, 2025), prior works have also explored alternative tokenization paradigms such as variable-wise (Shao et al., 2022; Liu et al., 2024a), frequency-wise (Yi et al., 2023), phase-wise tokenization (Niu et al., 2025), and various improvements to patch-wise tokenization (Huang et al., 2024; Abeywickrama et al., 2026). However, in the context of TSFMs, the design space of tokenization is constrained by the requirement to support arbitrary input lengths. As a result, many existing tokenization improvements are difficult to directly apply to TSFMs. In addition, point-wise tokenization is less favored due to its limited scalability under long-context settings. More recently, tokenization strategies specifically designed for TSFMs have been proposed, including domain-specific tokenization (Özgün Turgut et al., 2025) and wavelet-based tokenization (Masserano et al., 2025). Nevertheless, patch-wise tokenization remains the dominant paradigm in current TSFM architectures.

C. Supplementary Analysis

C.1. Limitations of Patch Tokenization

In this section, we further analyze the drawbacks of patch tokenization and provide empirical evidence to substantiate our claims.

Most existing pretrained time series models adopt patch tokenization. Although patching effectively increases the semantic density of tokens and reduces the total number of tokens (Nie et al., 2023), it also introduces several nontrivial limitations.

First, patching inherently entangles fine-grained temporal variations within a patch, which weakens the model’s ability to reason at the point level. This mismatch becomes particularly evident in tasks that require precise temporal localization, such as point-wise imputation and anomaly detection. Moreover, models pretrained with patch-level reconstruction objectives tend to overfit to patch-wise missing patterns, leading to degraded robustness when the corruption pattern shifts to point-level missingness. We provide empirical evidence for these limitations in Table 6. Notably, although point-wise missing constitutes a strictly simpler corruption pattern than patch-wise missing, the performance of MOMENT drops by over 20% when evaluated under point-missing imputation. Last but not least, patch tokenization struggles to capture high-frequency dynamics. Consider a sequence whose period equals the patch length: all patch tokens become identical, causing the Transformer to degenerate into a feed-forward MLP.

Table 6. Comparison of MOMENT’s performance between patch-level missing and point-level missing. Results are averaged over four masking ratios {12.5%, 25%, 37.5%, 50%}.

Datasets		ETTm1		ETTh2		Weather		Avg	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
MOMENT	Patch	0.226	0.284	0.133	0.239	0.071	0.102	0.143	0.208
	Point	0.275	0.335	0.166	0.276	0.083	0.142	0.175	0.251
	Δ	-21.7%	-18.0%	-24.8%	-15.5%	-16.9%	-28.2%	-22.4%	-20.7%

C.2. Detailed Efficiency Analysis

This section provides a detailed comparison of the computational complexity between ZEUS and a vanilla Transformer.

Baseline: Vanilla Transformer We consider a vanilla Transformer with point-wise tokenization and the same total number of layers as ZEUS. For a sequence of length L and hidden dimension d , the self-attention module incurs a computational cost of $\mathcal{O}(BL^2d)$ per layer, which dominates the overall FLOPs. With N layers, the total attention complexity is

$$\mathcal{C}_{\text{vanilla}} = \mathcal{O}(BNL^2d). \quad (10)$$

ZEUS ZEUS employs a multi-scale architecture with temporal scales $\{s_i\}$. The attention complexity at scale s_i is computed as Eq.(6).

Quantitative Comparison Using the configuration in our implementation, most layers in ZEUS operate on temporally downsampled representations (e.g., $L/8$ and $L/32$), while only a small number of layers attend at full resolution. Let $d = 384$ and $N = 12$, then we obtain

$$\mathcal{C}_{\text{ZEUS}} = B \cdot L^2 \left(3 \cdot 384 + \frac{6}{8^2} \cdot 768 + \frac{3}{32^2} \cdot 768 \right), \quad (11)$$

$$\mathcal{C}_{\text{vanilla}} = B \cdot L^2 \cdot 12 \cdot 384, \quad (12)$$

$$\mathcal{C}_{\text{ZEUS}} \approx 0.27 \mathcal{C}_{\text{vanilla}}, \quad (13)$$

which indicates that ZEUS achieves an approximately $3.8\times$ reduction in self-attention FLOPs compared to a vanilla Transformer with the same depth.

D. Experimental Details

D.1. Point Forecasting

Benchmarks We conduct experiments on six widely-recognized datasets in long-term forecasting benchmarks (Wu et al., 2023; Shao et al., 2024), including:

- **ETT** (Electricity Transformer Temperature) (Zhou et al., 2021) contains 7 features of electricity transformer data collected from two separate counties from July 2016 to July 2018. It contains four datasets: **ETTh1**, **ETTh2**, **ETTm1**, **ETTm2**, where ETTh1 and ETTh2 are recorded every hour, and ETTm1 and ETTm2 are recorded every 15 minutes.

- **ECL** (Electricity) (Wu et al., 2021) records the hourly electricity consumption data of 321 clients from 2012 to 2014. Each variable represents a client’s electricity consumption.
- **Weather** (Wu et al., 2021) includes 21 meteorological factors collected every 10 minutes from the weather station of the Max Planck Biogeochemistry Institute in 2020.

Following common practice, we consider four prediction horizons $\{96, 192, 336, 720\}$. For each horizon, the context length is selected via hyperparameter search from the set $\{512, 720, 1024, 2048, 3072\}$.

Baselines We compare our methods with three categories of baselines: (1) *TSFMs*: **MOMENT** (Goswami et al., 2024), **Timer** (Liu et al., 2024b), **UniTS** (Gao et al., 2024); (2) *Pretrained forecasting models*: **Kairos** (Feng et al., 2025), **Toto** (Cohen et al., 2025), **Sundial** (Liu et al., 2025b); (3) *Task-specific models*: **ModernTCN** (donghao & xue, 2024), **GPT4TS** (Zhou et al., 2023), **TimesNet** (Wu et al., 2023), **PatchTST** (Nie et al., 2023).

For baselines that include multiple model variants, we report results for the variant with the strongest overall performance (typically the largest ones). Subscripts $_b$ and $_l$ denote the base and large variants, respectively. Notably, the prediction head of MOMENT is randomly initialized and therefore requires fine-tuning before being applicable to forecasting, making it unsuitable for zero-shot evaluation. Consequently, we use its reconstruction head instead and formulate the forecasting task as an extrapolative reconstruction task, following the same setup used for ZEUS.

Metrics We adopt **Mean Squared Error (MSE)** and **Mean Absolute Error (MAE)** as evaluation metrics, where lower values indicate better performance. They are defined as follows.

$$\text{MSE}(\hat{y}, y) = \frac{1}{N \cdot T} \sum_{i=1}^N \sum_{t=1}^T (\hat{y}_t^i - y_t^i)^2, \quad (14)$$

$$\text{MAE}(\hat{y}, y) = \frac{1}{N \cdot T} \sum_{i=1}^N \sum_{t=1}^T |\hat{y}_t^i - y_t^i|. \quad (15)$$

Full Results The full results of point forecasting evaluation are provided in Table 7.

D.2. Probabilistic Forecasting

Benchmarks For probabilistic forecasting, we evaluate ZEUS’s performance on the **GIFT-Eval** benchmark (Aksu et al., 2024). GIFT-Eval comprises 23 datasets spanning multiple domains, including nature, energy, healthcare, finance, transportation, and cloud operations, covering 144,000 time series with 177 million data points. These datasets define a total of 97 forecasting configurations, including 55 short-term, 21 medium-term, and 21 long-term forecasting tasks, thereby providing a comprehensive assessment of the model’s predictive capabilities. Compared to long-term forecasting benchmarks used in point forecasting evaluation, GIFT-Eval places greater emphasis on short-term forecasting and additionally provides evaluation metrics for probabilistic predictions.

Baselines We include baselines: **Chronos-2** (Ansari et al., 2025), **TimesFM2.5** (Das et al., 2023), **TiRex** (Auer et al., 2025), **Xihe_{ultra}** (Sun et al., 2025), **FlowState** (Graf et al., 2025), **Kairos** (Feng et al., 2025), **Moirai2** (Woo et al., 2024), **Toto** (Cohen et al., 2025), **Sundial** (Liu et al., 2025b). The performance of baselines is taken from the official benchmark reports, using results available up to December 31, 2025.

Metrics Following the official protocol of GIFT-Eval, we report **Mean Absolute Scaled Error (MASE)** and **Continuous Ranked Probability Score (CRPS)** as evaluation metrics. Both metrics are lower-is-better.

MASE (Hyndman & Koehler, 2006) evaluates point forecasting performance by scaling the mean absolute error (MAE) of a model against that of a seasonal naive benchmark. It is defined as

$$\text{MASE} = \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{seasonal naive}}} \quad (16)$$

Table 7. Zero-shot point forecasting results across four prediction lengths {96, 192, 336, 720}. Datasets in pre-training are not evaluated on corresponding models, which are denoted by the dash (–). Best and second-best results are shown in **bold** and underlined, respectively.

Models	Time Series Foundation Models (Zero-shot)								Pretrained Forecasting Models (Zero-shot)								Task-Specific Models (Full-shot)										
	ZEUS		MOMENT		Timer		UniTS		Kairos _b		Toto _b		Sundial _l		Time-MoE _l		ChronosBolt _b		ModernTCN		GPT4TS		TimesNet		PatchTST		
	(Ours)		(2024)		(2024b)		(2024)		(2025)		(2025)		(2025b)		(2025)		(2024)		(2024)		(2023)		(2023)		(2023)		
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
ETTh1	96	0.341	0.372	0.662	0.548	0.437	0.425	0.423	0.437	0.396	0.385	0.382	0.381	0.346	0.383	0.350	0.382	0.420	0.388	0.381	0.401	0.388	0.404	0.408	0.426	0.382	0.403
	192	0.371	0.393	0.697	0.567	0.503	0.464	0.439	0.448	0.434	0.408	0.428	0.408	0.386	0.410	0.388	0.412	0.486	0.424	0.419	0.425	0.434	0.427	0.456	0.451	0.416	0.426
	336	0.384	0.403	0.700	0.574	0.510	0.470	0.460	0.452	0.452	0.418	0.457	0.422	0.410	0.426	0.411	0.430	0.517	0.445	0.443	0.441	0.463	0.446	0.508	0.496	0.443	0.444
	720	0.412	0.426	0.802	0.629	0.545	0.494	0.662	0.575	0.426	0.427	0.472	0.440	0.438	0.459	0.427	0.455	0.493	0.457	0.434	0.459	0.465	0.469	0.567	0.521	0.466	0.474
	Avg.	0.377	0.399	0.715	0.580	0.499	0.463	0.496	0.478	0.427	0.410	0.435	0.413	0.395	0.420	0.394	0.420	0.479	0.429	0.419	0.432	0.438	0.437	0.485	0.469	0.427	0.437
ETTh2	96	0.269	0.318	0.343	0.396	0.315	0.351	0.349	0.376	0.276	0.319	0.273	0.310	0.269	0.330	0.302	0.354	0.255	0.305	0.276	0.341	0.325	0.373	0.315	0.359	0.280	0.349
	192	0.319	0.358	0.372	0.414	0.428	0.417	0.423	0.420	0.353	0.369	0.339	0.356	0.325	0.373	0.364	0.385	0.333	0.355	0.340	0.384	0.402	0.425	0.435	0.422	0.351	0.393
	336	0.341	0.381	0.396	0.429	0.463	0.450	0.459	0.455	0.385	0.394	0.374	0.387	0.354	0.400	0.417	0.425	0.381	0.389	0.365	0.406	0.429	0.455	0.481	0.456	0.388	0.420
	720	0.352	0.398	0.464	0.473	0.447	0.457	0.478	0.472	0.386	0.412	0.375	0.400	0.389	0.443	0.537	0.496	0.393	0.408	0.401	0.435	0.464	0.480	0.455	0.461	0.424	0.445
	Avg.	0.320	0.364	0.394	0.428	0.413	0.419	0.427	0.431	0.350	0.374	0.340	0.363	0.334	0.387	0.405	0.415	0.341	0.364	0.346	0.392	0.405	0.433	0.422	0.425	0.361	0.402
ETTm1	96	0.272	0.324	0.610	0.517	0.690	0.526	0.643	0.507	0.295	0.322	0.320	0.333	0.273	0.329	0.309	0.357	0.303	0.311	0.283	0.340	0.298	0.349	0.358	0.373	0.278	0.330
	192	0.309	0.349	0.692	0.549	0.745	0.560	0.688	0.532	0.336	0.355	0.371	0.364	0.312	0.357	0.346	0.381	0.370	0.352	0.326	0.363	0.338	0.371	0.428	0.412	0.327	0.361
	336	0.334	0.368	0.753	0.567	0.949	0.637	0.704	0.547	0.367	0.380	0.408	0.388	0.343	0.378	0.373	0.408	0.410	0.382	0.363	0.384	0.370	0.391	0.494	0.447	0.363	0.388
	720	0.372	0.396	0.799	0.584	0.965	0.650	0.723	0.565	0.393	0.402	0.485	0.426	0.397	0.413	0.475	0.477	0.497	0.428	0.425	0.418	0.421	0.421	0.550	0.480	0.414	0.423
	Avg.	0.322	0.359	0.714	0.554	0.837	0.593	0.690	0.538	0.348	0.365	0.378	0.396	0.331	0.369	0.376	0.405	0.395	0.368	0.346	0.376	0.357	0.383	0.458	0.428	0.346	0.376
ETTm2	96	0.168	0.247	0.276	0.342	0.213	0.295	0.235	0.306	0.160	0.238	0.172	0.237	0.172	0.255	0.197	0.286	0.164	0.232	0.172	0.266	0.171	0.262	0.175	0.255	0.162	0.249
	192	0.222	0.287	0.322	0.368	0.306	0.348	0.291	0.338	0.220	0.283	0.232	0.280	0.227	0.296	0.250	0.322	0.233	0.280	0.226	0.299	0.244	0.310	0.251	0.305	0.221	0.291
	336	0.268	0.319	0.384	0.401	0.433	0.428	0.346	0.372	0.271	0.317	0.290	0.320	0.275	0.331	0.337	0.375	0.299	0.323	0.281	0.333	0.293	0.346	0.307	0.343	0.270	0.323
	720	0.336	0.366	0.453	0.441	0.539	0.482	0.441	0.433	0.357	0.373	0.372	0.375	0.343	0.378	0.480	0.461	0.414	0.391	0.380	0.391	0.391	0.407	0.412	0.407	0.369	0.386
	Avg.	0.249	0.305	0.359	0.388	0.373	0.388	0.328	0.362	0.252	0.303	0.267	0.303	0.254	0.315	0.258	0.315	0.278	0.307	0.265	0.322	0.275	0.331	0.286	0.328	0.256	0.312
ECL	96	0.126	0.213	0.734	0.685	0.193	0.232	0.349	0.417	-	-	0.129	0.213	0.130	0.227	-	-	-	-	0.131	0.227	0.136	0.233	0.184	0.289	0.139	0.235
	192	0.144	0.230	0.811	0.733	0.316	0.364	0.361	0.429	-	-	0.145	0.229	0.150	0.247	-	-	-	-	0.146	0.243	0.154	0.251	0.192	0.295	0.153	0.248
	336	0.161	0.248	0.859	0.755	0.342	0.383	0.430	0.483	-	-	0.163	0.247	0.170	0.268	-	-	-	-	0.166	0.264	0.172	0.270	0.193	0.299	0.168	0.267
	720	0.197	0.280	1.194	0.876	0.366	0.405	0.654	0.632	-	-	0.206	0.282	0.214	0.307	-	-	-	-	0.194	0.288	0.208	0.298	0.222	0.320	0.208	0.296
	Avg.	0.157	0.243	0.900	0.762	0.304	0.362	0.449	0.490	-	-	0.161	0.243	0.166	0.262	-	-	-	-	0.163	0.259	0.168	0.263	0.198	0.301	0.167	0.262
Weather	96	0.147	0.187	0.260	0.309	0.181	0.232	0.201	0.247	0.146	0.182	0.149	0.179	0.157	0.208	0.159	0.213	0.150	0.183	0.155	0.212	0.150	0.197	0.174	0.222	0.149	0.205
	192	0.190	0.229	0.298	0.336	0.284	0.326	0.260	0.294	0.192	0.228	0.192	0.223	0.207	0.256	0.215	0.266	0.197	0.230	0.196	0.245	0.197	0.241	0.229	0.265	0.195	0.248
	336	0.233	0.263	0.347	0.364	0.369	0.376	0.314	0.330	0.248	0.270	0.245	0.265	0.259	0.295	0.291	0.322	0.255	0.272	0.252	0.289	0.248	0.281	0.282	0.304	0.254	0.293
	720	0.297	0.308	0.400	0.401	0.469	0.432	0.390	0.392	0.338	0.330	0.310	0.312	0.327	0.342	0.415	0.400	0.344	0.329	0.323	0.334	0.326	0.334	0.360	0.352	0.346	0.354
	Avg.	0.217	0.247	0.326	0.353	0.326	0.342	0.291	0.306	0.231	0.253	0.224	0.245	0.238	0.275	0.256	0.288	0.237	0.254	0.232	0.270	0.230	0.263	0.261	0.286	0.236	0.275
# Wins	19	14	0	0	0	0	0	0	3	1	0	6	0	0	0	0	1	5	1	0	0	0	0	0	0	0	

with the Seasonal Naive MAE being defined as

$$\text{MAE}_{\text{seasonal naive}} = \frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}| \quad (17)$$

where T is the length of the *training* split of the series, y_t is the value of the series at time t , and m is the seasonal period.

CRPS (Gneiting & Raftery, 2007) is a scoring rule for probabilistic forecasting. Given a predicted distribution with CDF F and a ground truth value y , the CRPS is defined as:

$$\text{CRPS}(F, y) = \int_0^1 2\Lambda_\alpha(F^{-1}(\alpha), y) d\alpha, \quad (18)$$

where $\Lambda_\alpha(q, y)$ denotes the quantile loss.

In practice, CRPS is approximated by a discrete sum over a finite set of quantile levels. This approximation, often referred to as the mean weighted quantile loss (Park et al., 2021), is given by:

$$\text{CRPS} \approx \frac{1}{K} \sum_{k=1}^K \text{wQL}[\alpha_k], \quad (19)$$

where K is the number of quantile levels, and $\{\alpha_1, \alpha_2, \dots, \alpha_K\}$ are the selected quantile levels. The weighted quantile loss $\text{wQL}[\alpha]$ for each quantile level α is calculated as:

$$\text{wQL}[\alpha] = 2 \frac{\sum_t \Lambda_\alpha(\hat{q}_t(\alpha), y_t)}{\sum_t |y_t|}, \quad (20)$$

where $\hat{q}_t(\alpha)$ is the predicted α -quantile at time step t .

As in GIFT-Eval, both MASE and CRPS are further normalized by the performance of the Seasonal Naive forecast on the *test* split.

Full Results Following prior work (Auer et al., 2025; Liu et al., 2025b), we report the aggregated results in the main text, while the full results are provided in the project repository.

D.3. Imputation

Benchmarks Following prior work (Liu et al., 2024b; Zhang et al., 2025), we evaluate ZEUS on the same six datasets used for point forecasting. We mask $\{12.5\%, 25\%, 37.5\%, 50\%\}$ time steps in sequences of length 192.

Existing TSFMs are typically evaluated under patch-wise missing patterns. For example, MOMENT uses a patch size of 8 and therefore only considers missing segments of length 8 (Goswami et al., 2024), while Timer uses a patch size of 24 and considers missing segments of length 24 (Liu et al., 2024b). However, real-world time series exhibit more diverse missing behaviors, including both point-wise missing and contiguous segment missing with variable lengths (Yu et al., 2025b). To better reflect practical scenarios, we adopt two masking strategies:

1. **Random masking** simulates point-wise missing observations that occur sporadically in real-world time series. Such missing values often arise from packet loss during data transmission or occasional logging failures. This setting has been widely adopted in prior work (Wu et al., 2023; donghao & xue, 2024; Fu et al., 2026).
2. **Block masking**, where the lengths of contiguous missing segments are sampled from a geometric distribution with $p = 0.125$. This strategy reflects structured missing patterns in practice (e.g., sensor outages or system downtimes). The heavy-tailed nature of the geometric distribution aligns well with real-world missing behaviors, where short gaps are common but extended missing segments can still arise occasionally. We deliberately choose a small p to create more challenging missing patterns and to clearly distinguish this setting from random masking. The probability distribution of missing segment lengths is shown in Figure 9.

Baselines The baselines for the imputation task include TSFMs, namely **MOMENT** (Goswami et al., 2024), **Timer** (Liu et al., 2024b) and **UniTS** (Gao et al., 2024), as well as advanced task-specific models, including **GPT4TS** (Zhou et al., 2023), **ModernTCN** (donghao & xue, 2024), **TimesNet** (Wu et al., 2023), **PatchTST** (Nie et al., 2023), and **DLinear** (Zeng et al., 2023).

Metrics Consistent with the point forecasting setting, we adopt **Mean Squared Error (MSE)** and **Mean Absolute Error (MAE)** as evaluation metrics.

Full Results The full results of imputation evaluation are provided in Table D.3.

Table 8. Imputation results under random and block masking, where $\{12.5\%, 25\%, 37.5\%, 50\%\}$ of time steps are masked.

Models			Time Series Foundation Models (Zero-shot)								Task-Specific Models (Supervised)									
			ZEUS (Ours)		MOMENT (2024)		Timer (2024b)		UniTS (2024)		GPT4TS (2023)		ModernTCN (2024)		TimesNet (2023)		PatchTST (2023)		DLinear (2023)	
			MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Metrics			MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	Random	12.5%	0.064	0.159	0.300	0.352	0.429	0.423	0.672	0.548	0.071	0.183	0.067	0.178	0.059	0.163	0.099	0.209	0.111	0.233
		25%	0.071	0.167	0.351	0.382	0.464	0.441	0.761	0.602	0.089	0.201	0.077	0.193	0.078	0.187	0.119	0.230	0.147	0.267
		37.5%	0.082	0.179	0.406	0.412	0.502	0.460	0.861	0.635	0.110	0.223	0.091	0.209	0.098	0.208	0.140	0.248	0.181	0.295
		50%	0.099	0.195	0.472	0.445	0.541	0.478	0.856	0.671	0.141	0.253	0.110	0.226	0.119	0.232	0.165	0.268	0.217	0.322
		Avg.	0.079	0.175	0.382	0.398	0.484	0.451	0.788	0.614	0.103	0.215	0.086	0.202	0.089	0.198	0.131	0.239	0.167	0.279
	Block	12.5%	0.102	0.188	0.329	0.370	0.459	0.436	0.628	0.532	0.095	0.211	0.083	0.201	0.078	0.187	0.179	0.271	0.215	0.316
		25%	0.110	0.196	0.377	0.397	0.488	0.451	0.765	0.605	0.122	0.234	0.100	0.219	0.098	0.208	0.185	0.276	0.229	0.329
		37.5%	0.117	0.204	0.437	0.428	0.525	0.469	0.811	0.639	0.141	0.250	0.105	0.224	0.124	0.228	0.198	0.286	0.249	0.343
		50%	0.132	0.218	0.504	0.461	0.557	0.485	0.863	0.674	0.182	0.276	0.129	0.245	0.142	0.247	0.209	0.297	0.271	0.359
		Avg.	0.115	0.202	0.412	0.414	0.507	0.460	0.767	0.613	0.135	0.243	0.104	0.222	0.111	0.218	0.193	0.283	0.241	0.337
ETTh2	Random	12.5%	0.051	0.129	0.129	0.241	0.172	0.271	0.289	0.381	0.050	0.149	0.048	0.148	0.045	0.140	0.059	0.156	0.104	0.219
		25%	0.053	0.133	0.155	0.268	0.177	0.279	0.438	0.474	0.060	0.164	0.054	0.157	0.054	0.154	0.065	0.165	0.132	0.249
		37.5%	0.057	0.137	0.179	0.289	0.184	0.287	0.659	0.583	0.068	0.177	0.059	0.165	0.063	0.167	0.071	0.173	0.162	0.276
		50%	0.062	0.144	0.202	0.307	0.193	0.296	0.971	0.710	0.081	0.193	0.070	0.178	0.074	0.183	0.079	0.183	0.189	0.300
		Avg.	0.056	0.136	0.166	0.276	0.182	0.283	0.589	0.537	0.065	0.171	0.058	0.162	0.059	0.161	0.069	0.169	0.147	0.261
	Block	12.5%	0.063	0.145	0.169	0.268	0.184	0.280	0.326	0.401	0.070	0.179	0.067	0.177	0.067	0.172	0.088	0.193	0.244	0.330
		25%	0.064	0.147	0.183	0.286	0.187	0.286	0.470	0.488	0.076	0.188	0.072	0.184	0.074	0.181	0.090	0.194	0.275	0.348
		37.5%	0.068	0.152	0.199	0.302	0.194	0.293	0.683	0.591	0.085	0.199	0.074	0.186	0.080	0.189	0.094	0.199	0.288	0.361
		50%	0.073	0.159	0.218	0.319	0.201	0.300	0.991	0.715	0.095	0.210	0.079	0.192	0.089	0.198	0.099	0.206	0.304	0.374
		Avg.	0.067	0.151	0.192	0.294	0.192	0.290	0.618	0.549	0.082	0.194	0.073	0.185	0.078	0.185	0.093	0.198	0.278	0.353
ETTm1	Random	12.5%	0.034	0.110	0.185	0.275	0.755	0.523	0.617	0.513	0.046	0.144	0.041	0.136	0.037	0.127	0.046	0.140	0.056	0.163
		25%	0.036	0.113	0.242	0.316	0.688	0.508	0.717	0.579	0.060	0.165	0.047	0.146	0.047	0.143	0.056	0.154	0.074	0.189
		37.5%	0.039	0.117	0.303	0.355	0.644	0.499	0.763	0.615	0.069	0.175	0.052	0.153	0.056	0.157	0.063	0.164	0.094	0.213
		50%	0.044	0.126	0.371	0.392	0.616	0.494	0.821	0.653	0.083	0.190	0.065	0.173	0.073	0.179	0.068	0.171	0.120	0.239
		Avg.	0.038	0.116	0.275	0.335	0.676	0.506	0.730	0.590	0.065	0.169	0.051	0.152	0.053	0.152	0.058	0.157	0.086	0.201
	Block	12.5%	0.056	0.134	0.244	0.310	0.772	0.531	0.607	0.509	0.076	0.180	0.058	0.164	0.061	0.158	0.113	0.208	0.163	0.273
		25%	0.061	0.138	0.292	0.343	0.710	0.517	0.722	0.582	0.086	0.193	0.069	0.175	0.075	0.176	0.116	0.212	0.177	0.282
		37.5%	0.065	0.143	0.347	0.376	0.669	0.508	0.777	0.621	0.100	0.205	0.078	0.185	0.082	0.182	0.117	0.211	0.192	0.295
		50%	0.074	0.152	0.406	0.409	0.640	0.502	0.831	0.657	0.113	0.218	0.091	0.200	0.087	0.194	0.121	0.215	0.211	0.311
		Avg.	0.064	0.142	0.322	0.360	0.698	0.515	0.734	0.592	0.094	0.199	0.074	0.181	0.076	0.178	0.117	0.212	0.186	0.29
ETTm2	Random	12.5%	0.024	0.080	0.076	0.180	0.126	0.239	0.203	0.324	0.027	0.104	0.026	0.101	0.025	0.100	0.029	0.104	0.064	0.169
		25%	0.025	0.081	0.095	0.206	0.123	0.237	0.348	0.428	0.031	0.115	0.030	0.110	0.031	0.112	0.032	0.110	0.099	0.211
		37.5%	0.028	0.085	0.112	0.227	0.124	0.239	0.575	0.551	0.036	0.124	0.034	0.118	0.033	0.118	0.037	0.119	0.115	0.230
		50%	0.031	0.091	0.128	0.244	0.127	0.242	0.893	0.687	0.042	0.134	0.039	0.126	0.038	0.127	0.04	0.126	0.128	0.244
		Avg.	0.027	0.084	0.103	0.214	0.125	0.239	0.505	0.498	0.034	0.119	0.032	0.114	0.032	0.114	0.035	0.115	0.102	0.214
	Block	12.5%	0.032	0.095	0.115	0.213	0.133	0.245	0.249	0.353	0.044	0.139	0.043	0.136	0.040	0.131	0.049	0.138	0.170	0.268
		25%	0.033	0.097	0.117	0.225	0.130	0.243	0.382	0.446	0.045	0.141	0.044	0.138	0.043	0.134	0.050	0.140	0.183	0.286
		37.5%	0.035	0.100	0.127	0.240	0.130	0.244	0.602	0.561	0.048	0.147	0.045	0.140	0.046	0.140	0.053	0.144	0.198	0.300
		50%	0.038	0.105	0.140	0.255	0.132	0.246	0.913	0.692	0.053	0.153	0.052	0.149	0.050	0.146	0.056	0.149	0.212	0.312
		Avg.	0.035	0.099	0.125	0.233	0.131	0.245	0.537	0.513	0.048	0.145	0.046	0.141	0.045	0.138	0.052	0.143	0.191	0.292
ECL	Random	12.5%	0.037	0.120	0.206	0.335	0.309	0.413	0.729	0.689	0.104	0.225	0.071	0.188	0.097	0.217	0.068	0.183	0.076	0.193
		25%	0.041	0.126	0.262	0.388	0.369	0.467	0.926	0.790	0.111	0.233	0.093	0.219	0.102	0.221	0.080	0.200	0.100	0.226
		37.5%	0.047	0.135	0.330	0.444	0.443	0.528	0.935	0.797	0.117	0.238	0.114	0.244	0.106	0.225	0.099	0.224	0.122	0.252
		50%	0.056	0.148	0.417	0.510	0.527	0.588	0.960	0.806	0.123	0.245	0.138	0.269	0.110	0.229	0.113	0.240	0.147	0.278
		Avg.	0.045	0.132	0.304	0.419	0.412	0.499	0.888	0.771	0.114	0.235	0.104	0.230	0.104	0.223	0.090	0.212	0.111	0.237
	Block	12.5%	0.050	0.135	0.226	0.347	0.345	0.438	0.617	0.620	0.114	0.231	0.107	0.228	0.105	0.223	0.100	0.217	0.124	0.244
		25%	0.054	0.140	0.279	0.396	0.399	0.485	0.917	0.785	0.118	0.240	0.114	0.234	0.108	0.227	0.112	0.233	0.140	0.264
		37.5%	0.059	0.148	0.352	0.456	0.469	0.541	0.947	0.800	0.128	0.257	0.126	0.251	0.111	0.230	0.123	0.247	0.158	0.283
		50%	0.067	0.159	0.451	0.529	0.551	0.599	0.964	0.808	0.136	0.268	0.141	0.276	0.115	0.236	0.130	0.254	0.179	0.304
		Avg.	0.058	0.146	0.327	0.432	0.441	0.516	0.861	0.753	0.124	0.249	0.122	0.247	0.110	0.229	0.116	0.238	0.150	0.274
Weather	Random	12.5%	0.027	0.032	0.067	0.112	0.111	0.162	0.135	0.206	0.030	0.062	0.030	0.056	0.029	0.058	0.			

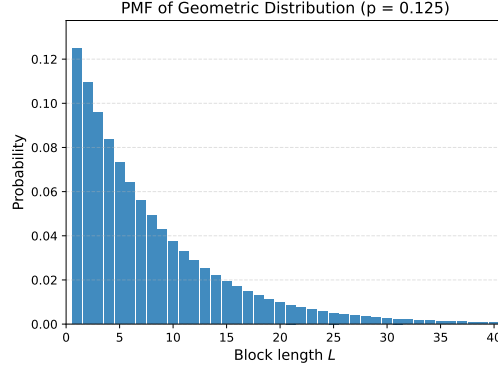


Figure 9. PMF of the geometric distribution used to sample missing segment lengths. The expected block length is 8, and with 99% probability the block length is smaller than 35.

D.4. Anomaly Detection

Benchmarks We evaluate the anomaly detection task on the UCR Anomaly Archive (Wu & Keogh, 2021), which consists of 250 tasks spanning diverse domains such as medicine, sports, entomology, and space science. The dataset contains both realistic and synthetic anomalies, addressing several limitations of earlier anomaly detection benchmarks. We conduct experiments on a subset of 42 tasks selected by MOMENT², which covers a wide range of domains and exhibits sufficient data diversity (Goswami et al., 2024).

Previous work typically evaluates models using a fixed window size. However, datasets in the UCR archive exhibit substantial variation in periodicity, making a fixed window potentially detrimental to model performance. Therefore, we treat the window size as a hyperparameter and select the optimal value via search from {64, 256, 512, 1024, 2048}.

Baselines The baselines for anomaly detection include *TSFMs*—MOMENT (Goswami et al., 2024), Timer (Liu et al., 2024b), and UniTS (Gao et al., 2024)—as well as *task-specific models*, including ModernTCN (donghao & xue, 2024), GPT4TS (Zhou et al., 2023), TimesNet (Wu et al., 2023), PatchTST (Nie et al., 2023), and Anomaly Transformer (Xu et al., 2022).

Metrics Following prior work (Wu et al., 2023; Goswami et al., 2024), we use the adjusted F1 score as the evaluation metric. Specifically, point adjustment is applied such that if an anomaly is detected at any time point within a ground-truth anomalous segment, the entire segment is considered correctly detected. Precision and recall are then computed based on these adjusted predictions, and the adjusted F1 score is calculated accordingly. Compared to the standard F1 score, the adjusted F1 score better reflects detection quality at the event level and reduces over-penalization caused by slight temporal misalignment.

Full Results Table 9 presents the full results of anomaly detection.

D.5. Classification

Benchmarks We conduct evaluations on the UEA classification archive (Bagnall et al., 2018), which comprises 30 multivariate time series classification datasets. Since four datasets (namely CharacterTrajectories, InsectWingBeat, JapaneseVowels, and SpokenArabicDigits) contain sequences with unequal lengths, some baselines are incompatible with them and are marked as NA in the corresponding works. To enable a fair comparison using dataset-averaged metrics, we conduct evaluations only on the 26 datasets with equal-length sequences.

While prior work on classification tasks relies on fine-tuning (Gao et al., 2024; Zhang et al., 2025) or linear probing (Goswami et al., 2024), ZEUS is evaluated in a tuning-free setting. Accordingly, we use a non-parametric 1-nearest neighbor

²Although MOMENT originally states that 45 tasks are selected, only 42 tasks are reported in its published results. We therefore follow the reported tables and conduct our evaluation on these 42 tasks.

Table 9. Adjusted best F1 score on 42 datasets sampled from the UCR Anomaly archive.

Models	TSFMs (Zero-shot)				Task-Specific Models (Supervised)				
	ZEUS (Ours)	MOMENT (2024)	Timer (2024b)	UniTS (2024)	ModernTCN (2024)	GPT4TS (2023)	TimesNet (2023)	PatchTST (2023)	Ano. Trans. (2022)
1sddb40	0.966	0.680	0.578	0.915	1.000	0.930	<u>0.972</u>	0.777	0.858
BIDMC1	1.000	1.000	1.000	0.858	0.998	1.000	1.000	1.000	<u>0.990</u>
CHARISfive	0.545	0.375	0.007	0.008	0.185	0.108	0.937	1.000	<u>0.968</u>
CHARISTen	0.882	0.504	0.310	0.667	<u>0.800</u>	0.352	0.382	0.795	0.144
CIMIS44AirTemperature3	1.000	0.350	1.000	<u>0.906</u>	0.361	0.180	<u>0.906</u>	<u>0.906</u>	0.085
CIMIS44AirTemperature5	1.000	0.615	0.889	1.000	1.000	0.200	<u>0.970</u>	0.897	0.390
ECG2	1.000	1.000	1.000	0.957	<u>0.990</u>	0.900	1.000	1.000	1.000
ECG3	0.962	0.823	0.195	0.562	0.823	0.840	<u>0.990</u>	0.995	0.360
Fantasia	0.984	0.943	0.989	0.968	0.997	0.870	0.990	<u>0.992</u>	0.971
GP711MarkerLFM5z4	1.000	0.871	0.535	0.843	0.959	0.819	<u>0.987</u>	1.000	0.930
GP711MarkerLFM5z5	1.000	1.000	1.000	0.963	0.889	0.929	0.950	1.000	0.852
InternalBleeding4	1.000	1.000	1.000	1.000	0.993	0.992	1.000	1.000	0.992
InternalBleeding5	1.000	1.000	1.000	1.000	1.000	0.994	1.000	1.000	0.940
Italianpowerdemand	<u>0.857</u>	0.525	0.080	0.990	0.445	0.141	0.353	0.630	0.010
Lab2Cmac011215EPG5	0.956	0.858	0.389	0.498	0.996	0.847	<u>0.990</u>	0.996	<u>0.990</u>
Lab2Cmac011215EPG6	<u>0.488</u>	0.020	0.071	0.127	0.340	0.100	0.151	0.381	0.806
MesoplodonDensirostris	1.000	0.967	1.000	1.000	1.000	1.000	1.000	<u>0.994</u>	1.000
PowerDemand1	0.953	0.809	0.956	0.955	0.887	0.760	<u>0.991</u>	0.997	0.870
TkeepFirstMARS	0.072	0.046	0.021	0.019	0.375	0.020	0.082	<u>0.226</u>	0.175
TkeepSecondMARS	1.000	1.000	0.153	1.000	0.645	0.417	0.741	1.000	<u>0.830</u>
WalkingAcceleration5	1.000	1.000	0.808	0.175	0.959	0.870	0.930	1.000	<u>0.990</u>
apneaecg	0.802	0.951	0.267	<u>0.997</u>	0.958	0.919	0.978	1.000	0.400
apneaecg2	0.952	0.990	1.000	<u>0.997</u>	0.990	1.000	0.980	1.000	0.817
gait1	0.940	0.793	0.418	0.714	<u>0.971</u>	0.739	0.715	0.955	1.000
gaitHunt1	0.849	0.500	0.045	0.721	<u>0.917</u>	0.579	0.840	0.969	0.080
insectEPG2	1.000	0.124	0.243	1.000	0.633	0.810	1.000	<u>0.943</u>	0.420
insectEPG4	0.962	<u>0.990</u>	0.187	0.123	0.610	0.210	1.000	1.000	0.980
ltstdbs30791AS	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
mit14046longtermecg	0.951	0.406	0.843	0.950	0.962	0.979	1.000	<u>0.993</u>	0.858
park3m	0.999	0.723	0.973	0.999	0.945	<u>0.987</u>	0.943	0.943	<u>0.987</u>
qtdbSel1005V	0.878	0.811	0.837	<u>0.945</u>	0.763	0.986	0.871	0.932	0.457
qtdbSel100MLII	0.891	0.883	0.888	<u>0.988</u>	0.970	0.996	0.929	0.958	0.809
respiration1	0.930	0.040	0.051	0.003	0.014	0.052	<u>0.920</u>	0.219	0.003
s20101mML2	1.000	0.687	0.995	0.995	1.000	<u>0.996</u>	1.000	1.000	1.000
sddb49	<u>0.998</u>	1.000	1.000	1.000	1.000	0.940	1.000	1.000	0.890
sel840mECG1	1.000	0.991	0.980	1.000	0.971	<u>0.999</u>	0.960	0.993	0.894
sel840mECG2	0.993	0.937	0.754	0.842	0.745	<u>0.983</u>	0.993	0.951	0.597
tilt12744mtable	0.244	0.206	0.079	0.159	0.128	<u>0.612</u>	0.734	0.003	0.070
tilt12754table	0.952	0.754	0.593	<u>0.906</u>	0.490	0.877	0.445	0.881	0.715
tiltAPB2	<u>0.990</u>	0.829	0.657	0.943	0.989	<u>0.990</u>	0.893	0.999	0.920
tiltAPB3	0.955	0.080	0.095	0.198	0.464	0.285	0.819	<u>0.921</u>	0.170
weallwalk	<u>0.857</u>	0.984	0.231	0.349	0.984	0.199	0.612	0.600	0.179
Avg. F1 Score	0.900	0.716	0.598	0.744	0.789	0.676	0.856	<u>0.877</u>	0.651
# Wins	21	10	10	11	12	6	13	<u>20</u>	6

(1-NN) classifier for evaluation, with optional PCA whitening applied as a feature normalization step.

Baselines The baselines include two *TSFMs*, **MOMENT** (Goswami et al., 2024) and **UniTS** (Gao et al., 2024), a *pretrained classification model* **VQ-shape** (Wen et al., 2024), and *task-specific models* including **ModernTCN** (donghao & xue, 2024), **SVP-T** (Zuo et al., 2023), **GPT4TS** (Zhou et al., 2023), **TimesNet** (Wu et al., 2023), **PatchTST** (Nie et al., 2023); and *classical methods* including **Rocket** (Dempster et al., 2020), **Shapelet Transformation with Random Forest (STRF)** (Hills et al., 2014), and **DTW** (Chen et al., 2013).

All pretrained baselines are fine-tuned in the evaluation of their original works and do not support a tuning-free setting. Specifically, MOMENT performs linear probing using an SVM classifier on pooled representations, VQ-Shape trains a linear head for downstream classification, and UniTS fine-tunes prompt tokens for task adaptation. To compare the tuning-free performance with other pretrained baselines, we replace the SVM classifier in MOMENT with an 1-NN classifier and re-evaluate the results, denoted as MOMENT[†].

Metrics Following prior work (Goswami et al., 2024), we use the accuracy as the evaluation metric.

Full Results The full results for classification tasks are provided in Table 10.

Table 10. Classification accuracy of methods across 26 UEA datasets. † denotes linear probing, and ‡ denotes prompt-based fine-tuning.

	TSFMs (1-NN)		Pretrained Models (Fine-tuned)				Task-Specific Models (Supervised)					Classical		
	ZEUS (Ours)	MOMENT (2024)	ZEUS [†] (Ours)	MOMENT [†] (2024)	UniTS [‡] (2024)	VQ-Shape [‡] (2024)	ModernTCN (2024)	SVP-T (2023)	GPT4TS (2023)	TimesNet (2023)	PatchTST (2023)	Rocket (2020)	STRF (2014)	DTW (2013)
ArticularyWordRecognition	0.977	0.987	0.990	0.990	0.927	0.987	0.983	0.993	0.933	0.973	0.927	0.996	0.917	0.987
AtrialFibrillation	0.467	0.067	0.533	0.200	0.133	0.520	0.467	0.400	0.333	0.333	0.333	0.249	0.267	0.200
BasicMotions	0.975	1.000	0.975	1.000	0.600	0.910	0.975	1.000	0.925	0.975	0.700	0.990	0.925	0.975
Cricket	0.986	0.903	1.000	0.986	0.958	0.978	0.958	1.000	0.847	0.903	0.889	1.000	0.944	1.000
DuckDuckGeese	0.500	0.400	0.560	0.600	0.320	0.360	0.560	0.700	0.560	0.580	0.220	0.461	0.380	0.600
EigenWorms	0.954	0.626	0.970	0.809	0.710	0.603	0.672	0.923	0.542	0.550	0.415	0.863	0.672	0.618
Epilepsy	0.986	0.971	1.000	0.993	0.942	0.893	0.957	0.986	0.855	0.877	0.913	0.991	0.978	0.964
ERing	0.926	0.944	0.959	0.963	0.830	0.960	0.952	0.937	0.948	0.927	0.937	0.981	0.889	0.133
EthanolConcentration	0.289	0.171	0.395	0.357	0.259	0.325	0.363	0.331	0.255	0.285	0.259	0.447	0.677	0.323
FaceDetection	0.552	0.509	0.639	0.633	0.549	0.653	0.708	0.512	0.656	0.677	0.668	0.694	0.567	0.529
FingerMovements	0.600	0.490	0.650	0.490	0.520	0.640	0.670	0.600	0.570	0.530	0.580	0.553	0.500	0.530
HandMovementDirection	0.365	0.216	0.419	0.324	0.365	0.546	0.527	0.392	0.473	0.595	0.514	0.446	0.419	0.231
Handwriting	0.319	0.218	0.294	0.308	0.137	0.270	0.306	0.433	0.327	0.311	0.251	0.567	0.104	0.286
Heartbeat	0.707	0.654	0.771	0.722	0.673	0.663	0.772	0.790	0.772	0.732	0.722	0.718	0.746	0.717
Libras	0.850	0.778	0.906	0.850	0.492	0.511	0.889	0.883	0.794	0.382	0.519	0.906	0.817	0.870
LSST	0.453	0.463	0.627	0.411	0.750	0.814	0.456	0.666	0.464	0.761	0.761	0.632	0.491	0.551
MotorImagery	0.610	0.530	0.620	0.500	0.540	0.680	0.560	0.650	0.500	0.610	0.600	0.530	0.510	0.500
NATOPS	0.767	0.767	0.844	0.828	0.756	0.810	0.917	0.906	0.917	0.833	0.756	0.885	0.794	0.883
PEMS-SF	0.786	0.792	0.919	0.896	0.844	0.865	0.891	0.867	0.873	0.844	0.809	0.856	0.925	0.711
PenDigits	0.936	0.957	0.948	0.972	0.894	0.973	0.973	0.983	0.974	0.984	0.974	0.996	0.855	0.977
PhonemeSpectra	0.166	0.180	0.261	0.233	0.119	0.087	0.131	0.176	0.113	0.146	0.081	0.284	0.155	0.151
RacketSports	0.836	0.704	0.803	0.796	0.684	0.851	0.816	0.842	0.770	0.855	0.757	0.928	0.842	0.803
SelfRegulationSCP1	0.698	0.669	0.785	0.840	0.795	0.904	0.934	0.884	0.915	0.908	0.795	0.866	0.846	0.775
SelfRegulationSCP2	0.556	0.483	0.578	0.478	0.528	0.596	0.603	0.600	0.517	0.539	0.506	0.514	0.489	0.539
StandWalkJump	0.530	0.333	0.600	0.400	0.400	0.787	0.467	0.467	0.333	0.533	0.467	0.456	0.467	0.200
UWaveGestureLibrary	0.769	0.916	0.878	0.909	0.838	0.888	0.867	0.941	0.844	0.863	0.828	0.944	0.762	0.903
Avg. Accuracy	0.675	0.605	0.728	0.672	0.599	0.695	0.707	<u>0.725</u>	0.654	0.673	0.601	0.704	0.635	0.609

E. Pretraining Dataset

We trained ZEUS on a diverse corpus of real and synthetic time series, comprising approximately 300B observations. We developed the Aegis-Syn synthetic dataset, enhancing KernelSynth dataset (Ansari et al., 2024) with additional diverse patterns. In training, synthetic data comprises roughly 10% of the sampled sequences. To mitigate the imbalance in pattern distributions across the training data, we adopt the balanced sampling strategy (Shao et al., 2025a) rather than assigning fixed probabilities per dataset. This promotes faster convergence and enhances model performance. To prevent data leakage, we exclude all evaluation datasets from pretraining data.

Table 11. Synthetic pattern components and their sampling distributions in Aegis-Syn.

Type	Pattern	Sampling Prob.	Typical Real-world Patterns
Seasonality	Sine	0.23	Smooth periodic behavior (e.g., temperature, seasonal demand)
	Triangle	0.13	Symmetric rise–fall cycles (e.g., tides, workload oscillations)
	Sawtooth	0.10	Gradual accumulation with abrupt reset (e.g., queues, buffers)
	Square	0.07	Binary on–off switching (e.g., machine states)
	Soft Square	0.03	Smoothed regime switching (e.g., controlled systems)
	Step	0.06	Piecewise-constant shifts (e.g., policy or tariff changes)
	Exponential Sawtooth	0.06	Accelerating growth before reset (e.g., congestion buildup)
	Half-rectified Sine	0.04	Positive-only activations (e.g., solar generation)
	Full-rectified Sine	0.03	Magnitude-only oscillations (e.g., vibration energy)
	Amplitude Modulation	0.05	Periodic signals with varying intensity (e.g., demand volatility)
	Frequency Modulation	0.05	Non-stationary periodicity (e.g., drifting rhythms)
Trend	Sparse Event (0-1)	0.08	Rare discrete events (e.g., faults, alarms)
	Pulse	0.07	Short-duration impulse signals (e.g., ECG, control inputs)
Trend	Linear	0.50	Long-term monotonic increase or decrease (e.g., growth, decay)
	Logistic	0.30	Saturating growth dynamics (e.g., adoption curves)
	Long-period Sinusoidal	0.20	Slow oscillatory trends (e.g., climate regimes)
Noise	Gaussian	0.60	Homoscedastic measurement noise (e.g., sensor and acquisition noise)
	ARMA	0.40	Temporally correlated noise (e.g., system inertia, residual dynamics)
Anomaly	None	0.50	Normal operating conditions
	Point Anomaly	0.20	Isolated spikes or drops (e.g., sensor glitches)
	Event Anomaly	0.15	Short-term level shifts (e.g., incidents, outages)
	Contextual Anomaly	0.10	Variance or noise regime changes (e.g., instability periods)
	Mixed Anomaly	0.05	Combination of multiple anomaly types

E.1. Real-World Datasets

The real-world data used in our study primarily come from two major sources. The detailed dataset composition is provided in Table 12 and 13.

1. **Chronos Datasets** (Ansari et al., 2024) contain 94B data points. We do not employ the TSMixup data augmentation technique proposed in Chronos.
2. **GiftEvalPretrain Datasets** (Aksu et al., 2024) is the pretraining dataset collection used in the GIFT-Eval benchmark. It comprises 71 univariate and 17 multivariate datasets spanning seven domains and 13 temporal frequencies, totaling 4.5 million time series and 230B data points. This dataset does not overlap with the test sets of the GIFT-Eval benchmark.

E.2. Aegis-Syn Dataset

We developed Aegis-Syn, a synthetic dataset designed to encompass diverse common patterns in time series data. Influenced by the pioneering KernelSynth (Ansari et al., 2024), the current mainstream approach for synthetic time series generation in model pretraining is based on Gaussian processes (GPs) (Shao et al., 2025a; Wang et al., 2025; Auer et al., 2025). While effective for modeling smooth and stationary dynamics, GPs are inherently ill-suited for representing non-smooth and discontinuous structures that frequently arise in practice, such as abrupt traffic surges or sudden drops in energy consumption.

To overcome this limitation, we construct a component bank that explicitly models diverse temporal structures commonly observed in real-world time series. The bank includes long-term trend components (e.g., linear drift and logistic growth), periodic patterns with both smooth and non-smooth waveforms (e.g., sinusoidal and square-like signals), stochastic noise components capturing random fluctuations, and anomaly components modeling irregular behaviors such as point anomalies and event-level deviations. The complete set of components and their corresponding sampling probabilities are summarized in Table 11. Based on this component bank, each synthetic time series is generated by sampling and composing components according to predefined probability distributions, following the generative procedure described in Algorithm 1.

Algorithm 1 Synthetic Time Series Generation in Aegis-Syn

```

1: INPUT: Sequence length  $T$ , component bank  $\mathcal{B}$ , sampling distributions  $\mathcal{P}$ 
2: OUTPUT: Synthetic time series  $\mathbf{x} \in \mathbb{R}^T$ 
3: Initialize  $\mathbf{x} \leftarrow \mathbf{0}$ 
4: // Trend sampling
5: Sample trend type  $b_{\text{trend}} \sim \mathcal{P}_{\text{trend}}$  from  $\mathcal{B}_{\text{trend}}$ 
6: Sample trend parameters  $\theta_{\text{trend}}$ 
7: Generate trend component  $\mathbf{x}_{\text{trend}} \leftarrow f(b_{\text{trend}}, \theta_{\text{trend}}, T)$ 
8:  $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{x}_{\text{trend}}$ 
9: // Periodic pattern sampling
10: Sample number of waveforms  $K \sim \mathcal{P}_K$ 
11: for  $k = 1, \dots, K$  do
12:   Sample waveform type  $b_k \sim \mathcal{P}_{\text{wave}}$  from  $\mathcal{B}_{\text{wave}}$ 
13:   Sample waveform parameters  $\theta_k = \{\omega_k, \phi_k, a_k\}$ 
14:   Generate periodic component  $\mathbf{x}_{\text{periodic}}^{(k)} \leftarrow f(b_k, \theta_k, T)$ 
15:    $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{x}_{\text{periodic}}^{(k)}$ 
16: end for
17: // Noise injection
18: Sample noise type  $b_{\text{noise}} \sim \mathcal{P}_{\text{noise}}$  from  $\mathcal{B}_{\text{noise}}$ 
19: Sample noise parameters  $\theta_{\text{noise}}$ 
20: Generate noise component  $\mathbf{x}_{\text{noise}} \leftarrow g(b_{\text{noise}}, \theta_{\text{noise}}, T)$ 
21:  $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{x}_{\text{noise}}$ 
22: // Anomaly injection
23: Sample anomaly type  $b_{\text{anom}} \sim \mathcal{P}_{\text{anomaly}}$  from  $\mathcal{B}_{\text{anomaly}}$ 
24: if  $b_{\text{anom}} \neq \text{NONE}$  then
25:   Sample anomaly parameters  $\theta_{\text{anom}}$ 
26:   Generate anomaly component  $\mathbf{x}_{\text{anom}} \leftarrow h(b_{\text{anom}}, \theta_{\text{anom}}, T)$ 
27:    $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{x}_{\text{anom}}$ 
28: end if
29: return  $\mathbf{x}$ 
    
```

F. Limitations and Future Work

Multivariate Correlations Like most pre-trained models and TSFMs, ZEUS focuses on univariate time series and handle multivariate time series with channel independent strategy, which has proven effective in practice, as evidenced by the experimental results and prior work such as PatchTST (Nie et al., 2023). Nevertheless, there exist scenarios where incorporating multivariate correlations can lead to significant improvements for downstream tasks. In that cases, adaptation techniques (e.g., CoRA (Qin et al., 2025)) can be adopted. Future work will explore explicitly incorporating multivariate correlations into this framework.

Generalization to Broader Tasks At present, ZEUS is primarily evaluated on five categories of downstream tasks that have been extensively explored in previous studies. On the one hand, some important downstream tasks, such as time series segmentation and causal discovery, are not yet well supported by the current framework. On the other hand, several compatible and promising tasks, such as irregular time series forecasting (forecasting tasks with structural missingness), remain unexplored in our evaluation. We leave these directions for future work.

Challenges in Classification Time series classification in a fully non-parametric setting remains challenging. While ZEUS with a 1-NN classifier shows promising performance, it is not yet able to outperform state-of-the-art baselines with full-shot training. This can be mainly attributed to two reasons. First, some UEA datasets contain extremely short (< 30) and high-dimensional (> 100) sequences, whereas ZEUS is better suited for long, univariate time series. Second, the label semantics in classification datasets are highly heterogeneous and strongly task-dependent. To address these issues, we explored contrastive learning (Chen et al., 2020) and SwAV-style prototype contrastive learning (Caron et al., 2020; Li et al., 2021), using joint training or post-training strategies to encourage more clustered representations in ZEUS. However, the

results were not satisfactory. We plan to continue this line of investigation in future work.

G. Case Study

Forecasting Showcases Figure 10 presents forecast examples from ZEUS. The first eight examples illustrate representative patterns from the Aegis-Syn dataset, while the remaining examples demonstrate the zero-shot forecasting capability of ZEUS. In all cases, ZEUS is able to produce accurate forecasts that closely follow the underlying temporal dynamics.

Reconstruction Showcases Figure 11 presents zero-shot reconstruction examples from ZEUS. For clarity, we primarily present the most challenging cases with a single large contiguous missing block. The first four examples are sampled from imputation benchmarks, demonstrating that the model can accurately reconstruct missing segments and preserve the underlying temporal patterns. The remaining four examples are from the UCR Anomaly Archive, showing that the model is able to precisely localize anomalous regions.

Clustering Analysis for Classification Figure 12 visualizes the representations of ZEUS on the UEA classification datasets after dimensionality reduction using t-SNE (van der Maaten & Hinton, 2008) and PCA. As can be observed, the t-SNE projections exhibit clear clustering patterns, while the PCA projections reveal a coherent low-dimensional manifold structure, indicating that ZEUS learns discriminative and structurally meaningful representations.

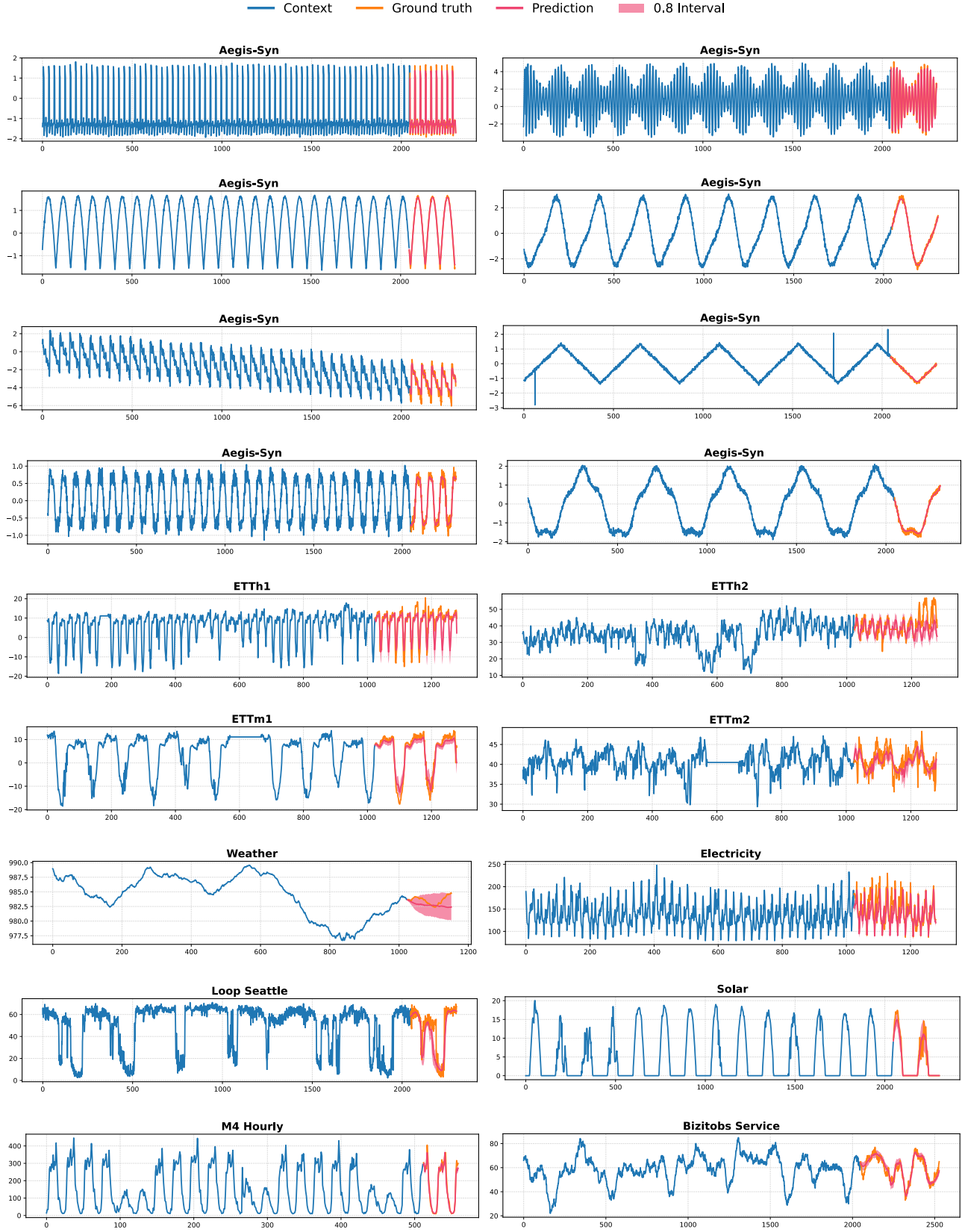


Figure 10. Example of forecasts from ZEUS.

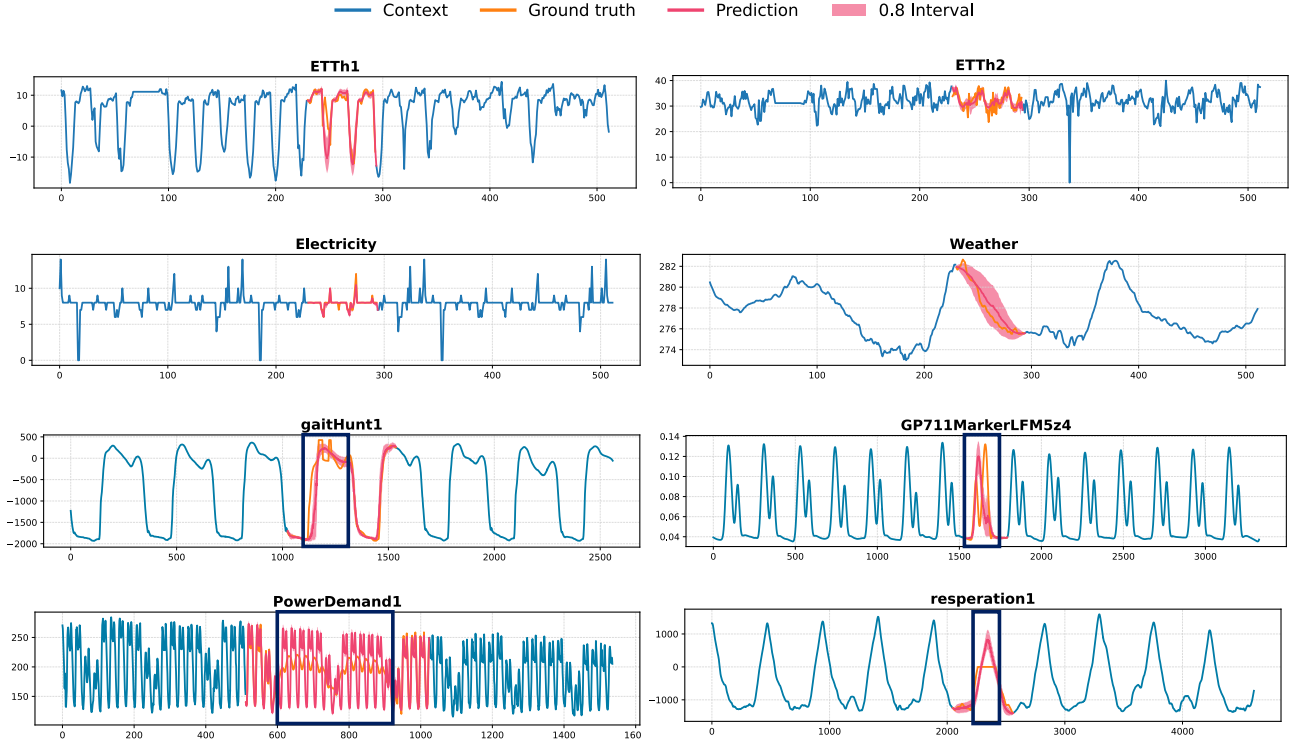


Figure 11. Zero-shot examples of reconstruction by ZEUS. Blue boxes denote the anomalies identified through reconstruction.

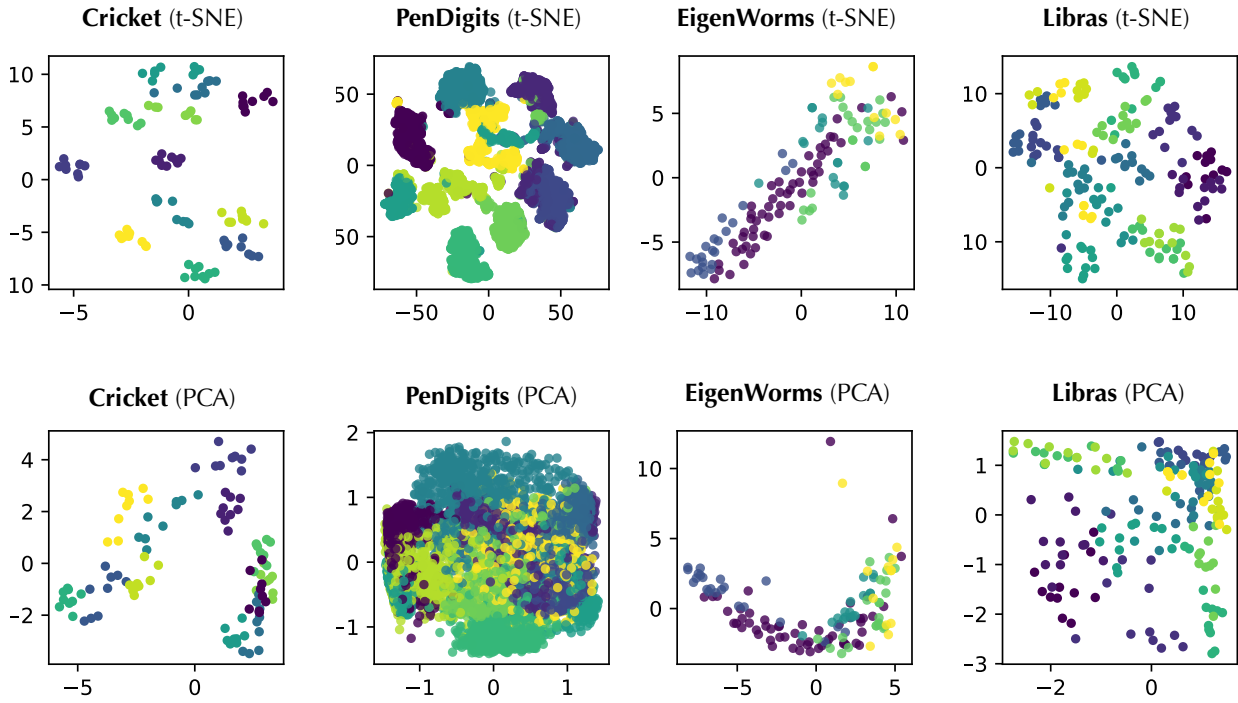


Figure 12. Visualization of representations learned by ZEUS on the UEA datasets.

Table 12. Statistics of the pretraining datasets.

Dataset	Domain	Frequency	# Time Series	# Observation
Mexico City Bikes	Transport	H	494	38,687,004
Solar	Energy	5T, H	5,166	1,085,664,720
Spanish Energy and Weather	Energy	H	1	35,064
Taxi	Transport	30T, H	4,856	3,346,350
USHCN	Climate	D	6,090	235,016,970
Weatherbench	Nature	H, D, W	225,280	79,375,265,520
Wiki (100k)	Web	D	100,000	274,100,000
Wind Farms	Energy	H, D	100,000	856,800,000
KDD Cup 2018	Energy	H	270	2,897,004
London Smart Meters	Energy	30T	5,560	166,528,896
M4	Econ/Fin	D, H, M, W	52,000	50,318,646
Pedestrian Counts	Transport	H	66	3,132,346
Rideshare	Transport	H	2,304	859,392
Temperature-Rain	Nature	D	32,072	22,290,040
Uber TLC	Transport	H, D	262	1,174,932
BDG-2 Panther	Energy	H	105	919,800
BDG-2 Fox	Energy	H	135	2,324,568
BDG-2 Rat	Energy	H	280	4,728,288
BDG-2 Bear	Energy	H	91	1,482,312
Low Carbon London	Energy	H	713	9,543,348
SMART	Energy	H	5	95,709
IDEAL	Energy	H	219	1,265,672
Sceaux	Energy	H	1	34,223
Borealis	Energy	H	15	83,269
Buildings900K	Energy	H	1,792,328	15,702,590,000
CMIP6	Climate	6H	1,351,680	1,973,453,000
ERA5	Climate	H	245,760	2,146,959,000
Azure VM Traces 2017	CloudOps	5T	159,472	885,522,908
Borg Cluster Data 2011	CloudOps	5T	143,386	537,552,854
Alibaba Cluster Trace 2018	CloudOps	5T	58,409	95,192,530
Taxi	Transport	30T	67,984	54,999,060
Wiki-Rolling	Web	D	47,675	40,619,100
M5	Sales	D	30,490	58,327,370
LargeST	Transport	5T	42,333	4,452,510,528
PEMS03	Transport	5T	358	9,382,464
PEMS04	Transport	5T	307	5,216,544
PEMS07	Transport	5T	883	24,921,792
PEMS08	Transport	5T	170	3,035,520
PEMS Bay	Transport	5T	325	16,937,700
Los-Loop	Transport	5T	207	7,094,304
Beijing Subway	Transport	30T	276	248,400
SHMetro	Transport	15T	288	1,934,208
HZMetro	Transport	15T	80	146,000
Q-Traffic	Transport	15T	45,148	264,386,688
Subseasonal	Climate	D	862	14,097,148
Subseasonal Precipitation	Climate	D	862	9,760,426

Table 13. Statistics of the pretraining datasets.

Dataset	Domain	Frequency	# Time Series	# Observation
Covid19 Energy	Energy	H	1	31,912
GEF12	Energy	H	20	788,280
GEF14	Energy	H	1	17,520
GEF17	Energy	H	8	140,352
PDB	Energy	H	1	17,520
BDG-2 Hog	Energy	H	24	421,056
BDG-2 Bull	Energy	H	41	719,304
BDG-2 Cockatoo	Energy	H	1	17,544
ELF	Energy	H	1	21,792
Wind Power	Energy	4S	1	7,397,147
Solar Power	Energy	4S	1	7,397,222
Oikolab Weather	Climate	H	8	800,456
Elecdemand	Energy	30T	1	17,520
Covid Mobility	Transport	D	362	148,602
Kaggle Web Traffic Weekly	Web	W	145,063	16,537,182
Extended Web Traffic	Web	D	145,063	370,926,091
M1 Yearly	Econ/Fin	Y	106	3,136
M1 Quarterly	Econ/Fin	Q	198	9,854
M1 Monthly	Econ/Fin	M	617	44,892
M3 Yearly	Econ/Fin	Y	645	18,319
M3 Quarterly	Econ/Fin	Q	756	37,004
M3 Monthly	Econ/Fin	M	1,428	141,858
M3 Other	Econ/Fin	Q	174	11,933
NN5 Daily	Econ/Fin	D	111	81,585
NN5 Weekly	Econ/Fin	W	111	11,655
Tourism	Econ/Fin	Y, Q, M	1212	150,822
CIF 2016	Econ/Fin	M	72	6,334
Traffic Weekly	Transport	W	862	82,752
Traffic Hourly	Transport	H	862	14,978,112
Australian Electricity Demand	Energy	30T	5	1,153,584
Sunspot	Nature	D	1	73,894
Vehicle Trips	Transport	D	329	32,512
Weather	Climate	D	3,010	42,941,700
FRED MD	Econ/Fin	M	107	76,612
Bitcoin	Econ/Fin	D	18	74,824
KDD Cup 2022	Energy	10T	134	4,727,519
GoDaddy	Econ/Fin	M	3,135	128,535
Favorita Sales	Sales	D	111,840	139,179,538
Favorita Transactions	Sales	D	54	84,408
China Air Quality	Nature	H	437	5,739,234
Beijing Air Quality	Nature	H	12	420,768
Residential Load Power	Energy	T	271	145,994,559
Residential PV Power	Energy	T	233	125,338,950
CDC Fluview ILINet	Healthcare	W	75	63,903
CDC Fluview WHO NREVSS	Healthcare	W	74	41,760
Project Tycho	Healthcare	W	1,258	1,377,707